The Missing "Loser's Curse": Experimental Evidence on Belief-Based Models in Common-Value Auctions*

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Abstract

Models based on mistaken beliefs, such as Cursed Equilibrium and Level-k Thinking, are among the leading explanations for the Winner's Curse observed in common-value auctions. We argue that these models predict a Loser's Curse in other auctions formats. Using an experiment with a within-subject design, we test for the presence of both curses in common-value uniform-price auctions. At the aggregate level, we find evidence of a strong Winner's Curse, but no evidence of a Loser's Curse. These aggregate findings cast some doubts on the ability of belief-based models to fully explain the Winner's Curse by themselves. Indeed, at the individual level, the behavior of most subjects is better described by Joy of Winning and Quantal Response Equilibrium. We also find suggestive evidence of failures of contingent thinking: subjects behave closer to the rational benchmark in a non-strategic task when the relevant contingency is made more salient.

JEL Classification: C91; C92; D44; D91.
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1 Introduction

Common-value auctions are those where bidders value an asset similarly but have partial and differing information (i.e., a signal) about its actual worth. A classic example is an auction for natural resources like oil or timber, but common-value elements can also appear in auctions for assets such as houses or paintings; in many such cases, bidders' valuations are shaped by their private preferences and the potential resale value, which introduces an aspect of shared valuation.

A large experimental literature shows that subjects tend to systematically overbid in common-value auctions. In particular, several studies have found evidence of the Winner's Curse: the winner bids so high that she ends up paying a price that exceeds the value of the item.[1] The existence and ubiquity of the Winner's Curse is one of the most robust findings in the experimental literature on auctions.[2] Moreover, the Winner's Curse can also manifest itself in bilateral-trade experiments, as shown by Samuelson and Bazerman (1985), Holt and Sherman (1994), and Charness and Levin (2009). This evidence, however, is at odds with the standard prediction of Bayes Nash equilibrium where buyers understand and correctly account for the adverse selection implied by winning an auction or by having an offer accepted.

Two popular explanations for the Winner's Curse are "Cursed Equilibrium" (Eyster and Rabin, 2005) and "Level-k Thinking" (Crawford and Iriberri, 2007). The first is an equilibrium model where a player underestimates the degree to which his opponents' actions are correlated with their private information; the latter is a non-equilibrium model of iterated best responses. In both models, bidders fail to fully account for the information revealed by winning an auction (i.e., that other bidders had lower signals), leading them to systematically overbid. However, the same logic predicts *underbidding* in multi-unit auctions. As Pesendorfer and Swinkels (1997) note, failing to condition on the auction's outcome generates a "Loser's Curse," in which bidders under-react to the positive information revealed by

[1] The first use of the term is by Capen et al. (1971) in the context of the early Outer Continental Shelf (OCS) oil lease auctions ("In fact [...] the Gulf has paid off at something less than the local credit union."). The term is often used to describe bidders suffering monetary losses, but a less extreme view is that the Winner's Curse results in lower than "normal" (or equilibrium) profits; see also Kagel and Levin (2002).

[2] See, for example, Bazerman and Samuelson (1983), Kagel and Levin (1986), Lind and Plott (1991), Levin et al. (1996), Avery and Kagel (1997), Ivanov et al. (2010), Levin et al. (2016), Koch and Penczynski (2018), and Charness et al. (2019). While we will discuss in detail the contributions most closely related to our paper, this literature is too voluminous for us to summarize here; hence, we refer the reader to the excellent survey by Kagel and Levin (2016).

losing, and may miss out on profitable trades they could have secured by bidding higher.[3]

This last observation is the key insight that motivates our study: belief-based explanations have a sharp cross-format implication when the primitives of the environment are held fixed. In a one-prize common-value auction, failing to condition on winning generates a Winner's Curse. In an otherwise identical multi-prize auction, where the pivotal contingency flips from winning to losing, the same failure to condition should instead generate a Loser's Curse. To test this implication, we conduct a within-subject experiment that compares bidding across these two formats. Specifically, subjects bid in *uniform-price* auctions with five bidders and homogeneous prizes, where the number of prizes varies. By varying the number of prizes for sale, we manipulate whether a bidder is exposed to the Winner's Curse or the Loser's Curse, holding fixed the information she observes about the prize's value and the number of competitors she faces. In particular, when there is only one prize for sale, buyers face a standard second-price auction in which the highest bidder wins and pays the second-highest bid; this is the environment in which the winner may suffer the Winner's Curse. With four prizes instead, the four highest bidders all get one prize each and pay the highest rejected bid; thus, the losing bidder in this auction could suffer from the Loser's Curse.

At an aggregate level, we observe a significant Winner's Curse in the one-prize auction, consistent with prior findings in the literature. While a sizable proportion of subjects tend to underbid in the four-prize auction, especially when they have a high signal, we do not find evidence for the Loser's Curse; we refer to this asymmetry as the missing Loser's Curse. The missing Loser's Curse is informative because belief-based models such as Cursed Equilibrium and Level-k Thinking typically predict that the same under-conditioning that generates a Winner's Curse in the one-prize auction should also generate a Loser's Curse in the four-prize auction, once the pivotal contingency flips from winning to losing.

A more recent literature suggests that individuals may suffer from a failure of contingent thinking; in a common-value auction, this entails that a bidder would fail to condition her bid on the hypothetical event that they win. This failure is not due to bidders having incorrect beliefs about the behavior of their competitors; rather, it may be due to individuals either being unable to, by themselves, conceive of the event of being pivotal or, even if they are able to construct it, fail to pay attention to it. Nevertheless, failures of contingent

[3]Similarly, buyers can be exposed to the Loser's Curse also in bilateral negotiations; in this context, Holt and Sherman (1994) find evidence of it, whereas Fudenberg and Peysakhovich (2016) do not.

thinking, like Cursed Equilibrium or Level-k, often also predict that subjects should experience the Loser's Curse in our four-prize auction. For experimental evidence of failures of contingent thinking, see Charness and Levin (2009), Esponda and Vespa (2014), Martínez-Marquina et al. (2019), Ngangoué and Weizsäcker (2021), Aina et al. (2024), Esponda and Vespa (2024) and Louis (2025); for an overview of the literature, see Niederle and Vespa (2023). Overall, these contributions show that experimental subjects tend to fare better when they are explicitly told about or put in the relevant contingency.

In order to evaluate the importance of failures of contingent thinking for common-value environments, subjects in our study also took part in a non-strategic valuation task. Keeping the common value of the prize and subjects' signals about it fixed, we informed them of how their signal ranked against others' and then elicited their willingness to pay for the prize. Notice that conditioning on the information about one's signal rank is an operation akin to the one that a bidder should perform in the symmetric Bayes Nash equilibrium of the auctions; yet, a subject's beliefs about how others bid are immaterial for this task. Our results indicate that subjects overbid for all signals, but their bids in this task are closer to the rational benchmark than in the auctions: the average absolute deviation from the rational benchmark in the valuation task is 84% of that in the four-prize auction and 69% of that in the one-prize auction. Hence, it appears that providing subjects with the relevant information to condition on helps them to some degree.

Going beyond the analysis of aggregate-level behavior, we find substantial heterogeneity at the individual level. Moreover, these differences in behavior between subjects are consistent: both the overall levels and the responsiveness of bids to signals show strong positive correlations across the two types of auction. Thus, there is a potential for behavioral models to explain behavior, and the best model might differ by subject.[4] We take a structural approach and estimate the parameters of several behavioral models for every subject; we then classify each subject as choosing according to the model with the lowest associated Akaike information criterion (AIC; Akaike, 1974). In addition to Cursed Equilibrium and Level-k Thinking, we consider two models that have been shown to explain deviations from equilibrium bidding in private-value auctions: Quantal Response Equilibrium (QRE) and the "Joy of Winning" hypothesis.[5] The majority of our subjects are classified as

[4]While several behavioral models can generate similar bid functions in common-value auctions, our experimental design mitigates this identification problem by leveraging within-subject comparisons across auction formats. This allows us to distinguish between mechanisms that would be observationally equivalent in the same auction format.

[5]Goeree et al. (2002) show that Quantal Response Equilibrium (QRE), in which players are more likely to

bidding as if they have a Joy of Winning, while a smaller proportion chooses according to QRE. Only a handful of subjects are classified as behaving according to Level-k Thinking or Cursed Equilibrium.

Overall, our findings suggest that models of limited strategic thinking (such as Cursed Equilibrium and Level-k) and failures of contingent thinking, while potentially capturing important aspects of why bidders deviate from Nash equilibrium in common-value auctions, do not by themselves account for the full cross-format pattern of behavior in our design. Indeed, in our data, we observe a generalized tendency to overbid, consistent with the Joy of Winning hypothesis, as well as significant noise, perhaps due to subjects' inability to correctly form conditional expectations or other forms of optimization errors, as suggested, for instance, by QRE. Importantly, our conclusions do not hinge on the literal interpretation of either the Joy of Winning model or QRE. Rather, we interpret them as reduced-form representations of broader classes of behavioral mechanisms that generate systematic overbidding and noise. Our central results are that belief-based explanations such as Level-k and Cursed Equilibrium struggle to jointly account for behavior in key parts of our design and that models that can successfully do so must capture both noisy decision-making and a general tendency to overbid.

The paper proceeds as follows. Section 2 describes our experimental design and the theoretical framework that motivates it. Section 3 presents our results at an aggregate level, discussing the extent to which they are consistent with the Winner's and Loser's Curses; this section also shows that behavior in the non-strategic task is closer to the rational benchmark. Section 4 presents the individual-level analysis, comparing several behavioral models according to how well they describe each individual's bidding behavior in uniform-price auctions. Section 5 provides a discussion of our findings and offers concluding remarks. The remainder of this section discusses the literature most closely related to our paper.

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choose actions with higher expected payoffs but may still make small errors, can explain overbidding in first-price auctions. In our setting, QRE predicts overbidding because bidding above one's signal is not punished sharply, so small stochastic deviations from best response can accumulate into systematic overbidding. Online Appendix B.3 shows the computed logit QRE for various logit parameter values. Cox et al. (1988) and Cooper and Fang (2008) show that the "Joy of Winning" (a preference for winning beyond material payoffs) can also rationalize overbidding in both first- and second-price auctions, as bidders may bid higher than the Nash Equilibrium bid to increase their chance of winning.

Related Literature

We contribute to the extensive experimental literature on the Winner's Curse in common-value auctions. Most of this literature has focused on first-price and second-price sealed-bid auctions, starting from the "Jar Experiment" of Bazerman and Samuelson (1983) who, using a first-price auction, auctioned the contents of four different jars full of coins to MBA students, with the common value being the sum of the coins in the jar; they found that although subjects tended to underestimate the value of the jar, the average winning bid was 25% higher than the true value. Since then, the canonical framework employed in this literature, which was first developed by Kagel and Levin (1986), has the common value being drawn from a uniform distribution. Subjects know the distribution from which the value is drawn, but they do not observe its realization; instead, they observe an unbiased iid signal of the value, which is also conditionally uniformly distributed.[6] Using the same framework, Levin et al. (1996) analyze English auctions, while Levin et al. (2016) focus on Dutch auctions. All of these studies document that bidders, including experienced ones, fall prey to the Winner's Curse.

Our study slightly departs from the literature mentioned above in two ways. First, for the common value we use the "Wallet Game" (Klemperer, 1998) formulation, where the value of the prize is given by the sum of all bidders' signals; for other experimental studies that employ this formulation, see Avery and Kagel (1997), Goeree and Offerman (2002), and Moser (2019).[7] Second, we consider uniform-price auctions of multiple, identical items. The experimental literature on uniform-price auctions has mostly focused on the case where bidders have private values and demand more than one unit; in this case, bidders have an incentive to reduce demand in an effort to obtain more favorable prices on the items they win (List and Lucking-Reiley, 2000; Kagel and Levin, 2001). By contrast, we focus on a common-value environment where bidders demand only one unit; we chose this setting in order to generate the opportunity for bidders to suffer from the Loser's Curse, while keeping the environment as similar as possible to that of a second-price auction.

The literature on the Loser's Curse is much less developed than the one studying the Winner's Curse. Yet, both curses stem from the same underlying mistake, namely a form

[6] For studies that employ this framework see Dyer et al. (1989), Lind and Plott (1991), Kagel and Richard (2001), Casari et al. (2007), Koch and Penczynski (2018), and Nagel et al. (2025).

[7] Yet another formulation also employed in the literature is the "Maximum Game" (Bulow and Klemperer, 2002), where the common value coincides with the maximum of the signals; for experimental studies that use this formulation, see Ivanov et al. (2010) and Camerer et al. (2016).

of selection neglect.[8] This point was first made by Holt and Sherman (1994) in the context of bilateral trading. In particular, they experimentally examined three versions of the "Acquiring-a-Company Game" of Samuelson and Bazerman (1985); in one of the versions, the distribution of the seller's types is such that if a buyer fails to properly condition on her offer being accepted, she will tend to bid too low, thereby failing to make a profitable acquisition.[9] Indeed, they find evidence of the Loser's Curse, with subjects bidding 20% below the rational benchmark; yet, using a similar design, Fudenberg and Peysakhovich (2016) instead find no evidence of underbidding, with subjects behaving close to the rational benchmark. As pointed out by Pesendorfer and Swinkels (1997), in uniform-price common-value auctions, bidders are simultaneously exposed to both curses; and a rational bidder should be able to avoid both. To the best of our knowledge, we are the first to design an experiment aimed at testing for the presence of the Loser's Curse in common-value auctions.

We also contribute to the recent literature on failures of contingent thinking. Roughly speaking, a failure of contingent thinking arises when agents are able to select the optimal action when a problem is presented in a way that helps them focus on all relevant contingencies, but fail to optimize if the problem is presented without such aids.[10] Failures of contingent thinking have been observed in voting games (Esponda and Vespa, 2014), Ellsberg-type and Allais-type decision problems (Esponda and Vespa, 2024), public-good contribution games (Calford and Cason, 2024), and in "Acquiring-a-Company" problems (Charness and Levin, 2009; Martínez-Marquina et al., 2019). More relevant for our study, Koch and Penczynski (2018), Moser (2019), and Nagel et al. (2025) find evidence of failures of contingent thinking in common-value auctions; however, there are several differences between their designs and ours. Koch and Penczynski (2018) compare bidding between a first-price auction and a transformed version of this auction that does not require conditional reasoning, and find that overbidding decreases significantly in the latter. Moser (2019) considers second-price auctions in which bidders are offered the opportunity to change their bid after learning whether it was the winning one; he finds that indeed subjects revise their bids, although not always for the better. Similarly, Nagel et al. (2025) find

[8] Massey and Thaler (2013) use the term when describing how top NFL draft picks are significantly overvalued in a manner that is inconsistent with rational expectations and efficient markets; yet, the mechanism we highlight is different.

[9] Holt and Sherman (1994) describe such bidder as being naïve; this is equivalent to a fully cursed bidder in the language of Eyster and Rabin (2005).

[10] A related notion is that of an obviously strategy-proof mechanism introduced by Li (2017).

that informing subjects in a first-price auction that they have the highest signal leads them to revise their bids downward; yet, such announcement would be immaterial if bidders were already conditioning on winning.[11] In contrast to these studies, we test for failures of contingent thinking using a non-strategic task; we find that subjects' decisions improve when they are provided with the relevant information to condition on, but they are still far from the rational benchmark.[12]

Finally, a recent experimental literature studies inference failures that arise when information is endogenously selected or when decisions are correlated across agents. In addition to auctions and bilateral trade, Bochet and Magnani (2024) document an "acceptance curse" in a matching market with private information and show that subjects do not adjust to a manipulation that changes the informativeness of acceptance, emphasizing the role of out-of-equilibrium beliefs. Similarly, Rees-Jones et al. (2024) document correlation neglect in student-to-school matching, showing that participants do not sufficiently account for correlation in admissions decisions. Relatedly, Barron et al. (2024) study selection neglect in investment decisions when agents learn from others' realized outcomes.

2 Experimental Design

In this section, we describe the details of our experiment. Section 2.1 outlines the theoretical framework that underpins our experimental design, together with the predictions of the symmetric Bayes Nash equilibrium benchmark. Section 2.2 describes our experimental implementation.

2.1 Theoretical Framework

There are $N \geq 2$ bidders, labeled $i = 1, ..., N$, and K identical items, with $N > K \geq 1$. Each bidder i privately observes a signal $s_i \in [\underline{s}, \overline{s}] \subset \mathbb{R}_+$; signals are independently drawn from the same distribution $F : [\underline{s}, \overline{s}] \to [0, 1]$ which admits a smooth strictly positive density f. We often refer to a bidder's signal as her type. Each bidder wants at most one item. The value of an item is the same for (but unknown to) all bidders, and is given by

[11] Nagel et al. (2025) have a multi-part experimental design aimed at identifying the key driving factors of the Winner's Curse in first-price auctions; because of the close connection to our own study, we defer the comparison between our and their results to Section 5.

[12] Similarly, Ngangoué and Schotter (2023) directly elicit subjects' estimates of the common-value item contingent on having the highest signal; we elicit these estimates for all signal ranks.

$$V(s_1, ..., s_N) = \sum_{i=1}^{N} s_i. \tag{1}$$

In our experiment, we are going to focus on so-called uniform-price auctions. In such auctions, bidders simultaneously submit sealed bids and the winning bidders are all charged the amount of the highest rejected bid. Hence, if there are K items, the K highest bidders each receive an item and pay a price p given by the $K+1$st highest bid; this procedure generalizes the single-object second-price auction. A bidder's payoff is equal to $V - p$ if she obtains an item and to zero otherwise.[13]

The distribution of the types of the other bidders is of central importance to any given bidder. Hence, taking the point of view of a generic bidder, let $y_1 > y_2 > ...y_{N-1}$ denote the order statistics of the types of her competitors. Then, as shown by Pesendorfer and Swinkels (1997), the unique symmetric Bayes Nash equilibrium strategy in a uniform-price auction for K items is given by

$$\beta_K^*(s_i) = \mathbb{E}[V|y_K = s_i, s_i]. \tag{2}$$

In words, bidder i submits a bid equal to the expected value of an item conditional on (i) her signal, and (ii) tying with her opponent with the Kth highest signal.

Finally, we introduce two additional functions that will play an important role in our experimental design. First, we let $\mathbb{E}[V|s_i]$ denote bidder i's expectation of the value of the item conditional on just her own signal. Next, we define bidder i's willingness to pay conditional on her signal and on knowing how her signal ranks compared to those of the other bidders as

$$\beta_{RV}(s_i, r) = \mathbb{E}[V|s_i, r] \tag{3}$$

where $r \in \{1, 2, ..., N\}$ denotes the rank of bidder i's signal; e.g., $r = 1$ means that bidder i has the highest signal.

2.2 Experiment

We conducted a laboratory experiment based on the setting described in Section 2.1, using a within-subject design with three treatments. In the first treatment, subjects participated

[13] Thus, we assume that bidders are risk neutral; yet, the analysis would be qualitatively similar under other risk attitudes so long as these are homogeneous across bidders.

in a one-prize ($K = 1$) second-price auction. In the second treatment, they participated in a four-prize ($K = 4$) fifth-price auction. The third treatment was a non-strategic valuation task. In each round, subjects completed all three treatments.

Each auction involved $N = 5$ buyers, and the common value of the prize followed equation (1), with private signals independently drawn from a uniform distribution over $[0, 5]$. In this setting, expression (2) becomes

$$\beta_1^*(s_i) = 3.5 s_i \tag{4}$$

for $K = 1$, and

$$\beta_4^*(s_i) = 7.5 + 3.5 s_i \tag{5}$$

for $K = 4$. Therefore, in our experimental implementation, the equilibrium bid in the four-prize auction has the same slope as that of the one-prize auction, but it is shifted upward by 7.5. Moreover, there is an underlying equivalence between these two auctions: The difference in payoffs for player i, $i = 1, ..., 5$, from bidding b_i or b_i' in the one-prize auction, given signals $s_1, \ldots, s_5$ and the bids $\{b_j : j \neq i\}$ of the other players, is the same as the difference in payoffs from bidding $25 - b_i$ or $25 - b_i'$ in the four-prize auction, given the transformed signals $5 - s_1, \ldots, 5 - s_5$ and transformed opponent bids $\{25 - b_j : j \neq i\}$. Given that the distribution of s_i is symmetric, this implies that an equilibrium in one auction can be used to recover an equilibrium in the other.[14]

In the third treatment, subjects completed a non-strategic valuation task in which, after being informed of their signal's rank, they had to indicate their maximum willingness to pay to acquire the same common-value prize. In theory, this willingness to pay should coincide with expression (3), which in this setting is given by

$$\beta_{RV}(s_i, r) = 3 s_i + 2.5(r - 1) \tag{6}$$

for $r = 1, 2, ..., 5$. Importantly, notice that

$$\beta_{RV}(s_i, 2) \geq \beta_1^*(s_i) > \beta_{RV}(s_i, 1)$$

and

$$\beta_{RV}(s_i, 5) \geq \beta_4^*(s_i) > \beta_{RV}(s_i, 4).$$

[14] We thank Joel Sobel for suggesting the existence of this transformation.

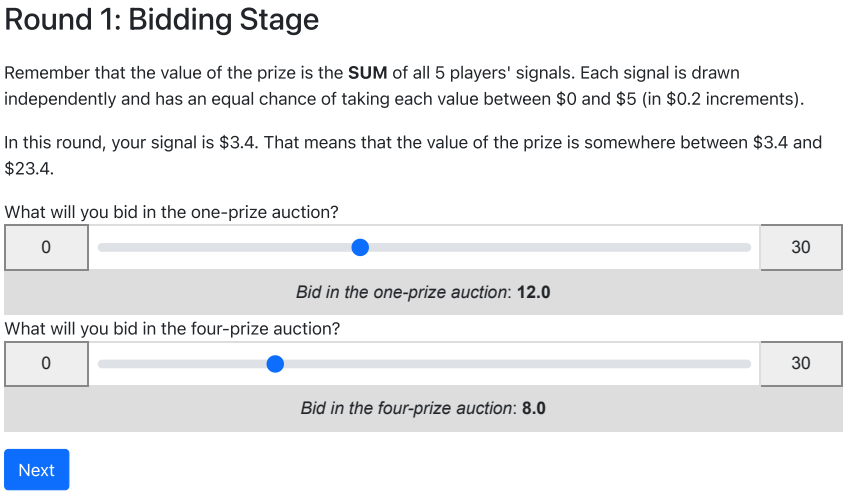


Figure 1: Bidding Stage

These inequalities provide us with meaningful bounds that allow us to check whether, when bidding in the auctions, subjects condition on the relevant contingency (provided that they choose correctly in the valuation task).

Each session began with the experimenter reading the instructions aloud; subjects were also presented with the same instructions both on paper and on their screen.[15] After all instructions had been read, subjects were presented with a detailed example showing a group of five participants, their signals, their choices in each task, and the associated outcomes. The subjects then completed a quiz to test their understanding of each task. After all subjects completed the quiz, they went on to complete 20 rounds of the experiment; each round was broken into two stages: the **bidding stage**, where subjects made bids for both auctions, and the **purchase stage**, where subjects completed the valuation task. Groups were rematched randomly after each round. At the end of the experiment, one decision within one round was chosen at random to be paid. Sessions lasted between 100 and 120 minutes, of which 10-15 minutes were spent reading instructions; on average, subjects earned $52.55.[16] The experiments were completed in-person through computer terminals, and all experiments were coded using oTree (Chen et al., 2016).

In the **bidding stage**, subjects were reminded of the structure of the common value and of the distribution of the signals. Then they were informed of the realization of their own signal. On a single screen, subjects made bids in both the one-prize auction and the four-

[15] Screenshots of instructions and all parts of the experiment can be found in Online Appendix E.
[16] Throughout the paper, the symbol $ denotes Australian dollars.

prize auction using sliders that were initially unselected and had to be clicked to activate.[17] Bids could be made in \$0.20 increments. Once both bids were entered, subjects clicked next to move to the next stage. An example of a bidding stage interface can be found in Figure 1.

In the **purchase stage**, subjects were again reminded of the structure of the common value and of the distribution of the signals, as well as the realization of their own signal; then, they were told the *rank* of their signal, as it compared to the other members of their group.[18,19] After that, subjects were asked to provide the maximum price they would be willing to pay to receive the prize.[20] This price was reported using a slider, which was not initialized and had to be clicked to be activated. The slider moved in \$0.20 increments. This task was incentivized using a Becker-Degroot-Marschack mechanism (BDM; Becker et al., 1964), whereby a random price was drawn uniformly on the interval from \$0.20 to \$30. If the price was below the reported maximum price, then the subject won the prize and paid the random price.[21] An example of a purchase stage interface can be found in Figure 2.

In both the bidding stage and the purchase stage, subjects were given a "budget" of \$30 out of which they made their bids or their purchasing choices. This was implemented to avoid losses. On the bottom of each decision page, subjects were presented with a short reminder of the instructions that were relevant for their decisions.

After completing both the bidding stage and the purchase stage, subjects were informed of the outcome of each task within the round. Specifically, they were informed of the overall value of the prize, the second-highest bid in the one-prize auction, the lowest bid

[17] Beyond limiting order effects, which we address in Appendix A.5, eliciting bids for both formats on the same screen keeps the information set, attention state, and interface context constant across decisions. It also lowers the risk that subjects confuse which bid corresponds to which format.

[18] The purchase stage *always* followed the bidding stage, as providing rank information would have altered strategic incentives in the auctions. In Appendix A.5, we examine whether this fixed ordering affected behavior and find no evidence that it significantly influenced the results.

[19] A fully rational decision maker should be able to update the possible range of the common value based on the additional information provided by the rank. We did not do this computation for the subjects exactly because we are interested in testing their ability to correctly update.

[20] Therefore, this task is in the spirit of, and generalizes, that of Ngangoué and Schotter (2023) who elicited subjects' estimates of a common-value lottery's expected payoff in the hypothetical event of having the highest signal.

[21] As a decision aid, the interface of the purchase stage also included an interactive feature helping subjects to understand at what prices they would purchase. Each purchase stage had a table containing values between \$0.20 and \$30 in \$0.20 increments. After activating the slider, the table was filled in with either an ✗ or a ✓ next to each price, indicating to subjects at what prices they would purchase. An example can be seen in Figure 26 of Online Appendix E.

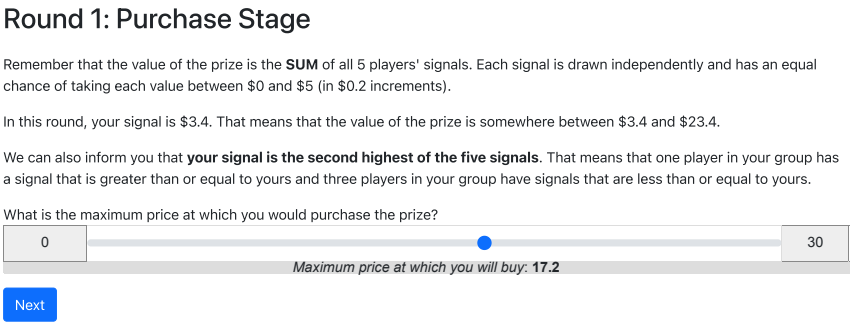


Figure 2: Purchase Stage

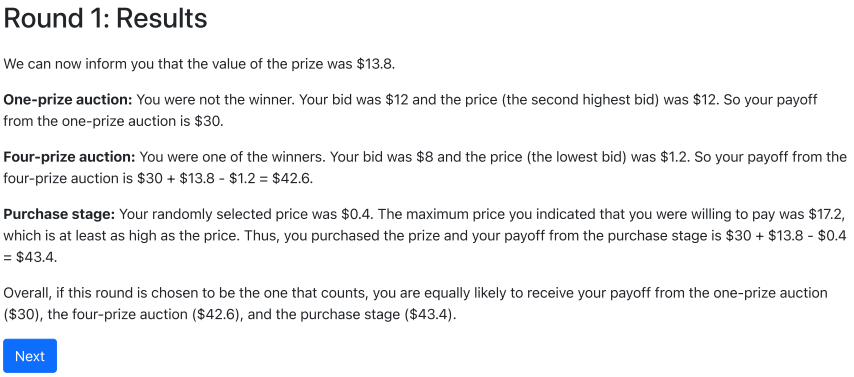


Figure 3: Feedback

in the four-prize auction, and the randomly drawn price.[22] They were informed of their payoffs from each decision task, and reminded that if this round was selected for payment, each task was equally likely to be chosen to be the one that counted for payments. An example of the feedback given can be found in Figure 3.

After subjects completed all 20 rounds, they were informed of their payments from the experiment and completed a short survey. They reported basic demographic information, completed a Cognitive Reflection Test (CRT; Frederick, 2005), and gave feedback about the experiment.

Overall, 120 subjects participated in our study, split across five experimental sessions that took place at the experimental laboratory of the Centre for Unified Behavioural and

[22]There are a variety of approaches regarding feedback in experimental common-value auctions. At one end of the spectrum, Garvin and Kagel (1994) and Casari et al. (2007) show all bids and their associated signals at the end of each round. On the other hand, Ivanov et al. (2010), Camerer et al. (2016) and Moser (2019) provide no feedback at all. Our approach is more balanced and resembles that of Ngangoué and Schotter (2023).

Economic Sciences (CUBES) of the University of Queensland in March 2024. Summary statistics of subject demographics can be found in Table 4 of Appendix A.1.

3 Aggregate Results

We start our analysis by describing our results at the aggregate level. Section 3.1 discusses the extent to which both the Winner's and Loser's Curses are present in our auction data. Section 3.2 compares the bidding patterns in our experiment with the predictions of the symmetric Bayes Nash equilibrium; moreover, we also show that experience led subjects in our experiment to bid slightly closer to the Nash equilibrium, but only in the one-prize auction. Finally, Section 3.3 presents the results of the non-strategic task, showing that subjects' choices seem to improve when they are directly provided with the relevant information on which to condition their bids. We preregistered the empirical specifications reported in Table 1 (analyzing how choices vary with private information across the three tasks) and Table 6. The other analyses were not preregistered and are therefore exploratory in nature.

3.1 Payoffs: Winner's and Loser's Curse

We begin by asking two simple but crucial questions about the payoffs in these auctions. First, do winners of one-prize auctions fall prey to the Winner's Curse? And, second, do losers of the four-prize auctions suffer from the Loser's Curse?

Many previous papers have found evidence of the Winner's Curse in common-value auctions (Kagel and Levin, 1986; Avery and Kagel, 1997; Goeree and Offerman, 2002; Ivanov et al., 2010; Camerer et al., 2016; Levin et al., 2016). We say that a winner is subject to the Winner's Curse if their empirical payoffs are lower than what they would have received had they bid lower and lost the auction. Rational bidders should not, on average, lose money from winning. However, if all bidders ignored the information conveyed by winning the auction and bid the expected value of the prize conditional on only their signal, winners would earn $29.17, losing 83 cents on average.

Result 1. Winners of the one-prize auctions are subject to the Winner's Curse on average.

Figure 4 shows the average payoffs that winners of the one-prize auctions received as compared to the payoffs that they would have received had they lost the auction. Winners

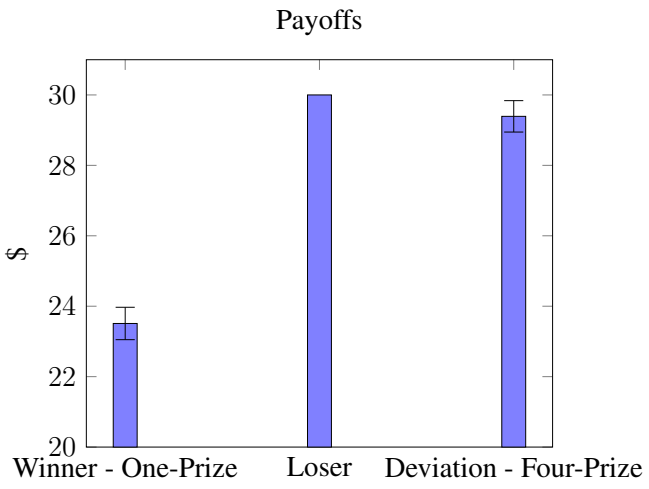


Figure 4: Cursed Payoffs. "Winner - One-Prize" shows the average payoffs of winners in the one-prize auction, and "Deviation - Four-Prize" shows the payoffs that losers of the four-prize auction would have received had they bid high enough to win. Winners of the one-prize auction would earn higher payoffs on average if they had bid zero and lost, but losers of the four-prize auction *would not* earn higher payoffs if they had maximized their bid and guaranteed themselves a win.

of the one-prize auctions receive less than \$24 on average, and this average is statistically different from \$30 ($p < 0.01$). In contrast, winners in the four-prize auction are *not* subject to the Winner's Curse, as they receive \$33.78 on average.[23]

As discussed in the Introduction, the classical argument that leads to the Winner's Curse in the one-prize auction also suggests the existence of a Loser's Curse in the four-prize auction. In particular, the Loser's Curse would imply that buyers who lose the auction would have been better off if they had deviated to a higher bid, winning the auction at a price equal to the next lowest bid. We say that such a loser is subject to the Loser's Curse. This gives a natural empirical measure of the Loser's Curse: In each four-prize auction, we compute the payoff a buyer would receive if they won the auction at a price equal to the fourth-highest bid and refer to this as the "Deviation". If subjects behaved consistently with Nash Equilibrium, the expected payoffs for the deviation would be \$29.17–less than they would receive from losing. If instead subjects bid the expected value of the prize

[23]We could also define a "Weak Winner's Curse" as winners receiving lower payoffs than what are predicted by Nash Equilibrium. Because our data shows evidence of the Winner's Curse holding on average in the one-prize auctions, it necessarily also shows evidence for this Weak Winner's Curse to hold on average. Winners in the four-prize auctions are *not* subject to the Weak Winner's Curse because the predicted payoffs for winners under Nash equilibrium in the four-prize auctions is \$32.08.

conditional on only their signal, deviation payoffs would be \$30.83.

Result 2. Losers of the four-prize auction are not subject to the Loser's Curse.

The evidence for Result 2 can again be found in Figure 4. The payoff associated with the deviation is *not* above the payoff that buyers receive from losing in these auctions, indicating that there is no evidence of a Loser's Curse.[24,25]

3.2 Bidding and Experience

Figure 5 shows average bids along with their 95% confidence intervals for each signal and both types of auction, comparing them to the BNE benchmark. While the confidence intervals for the average bids overlap for all signals, Table 5 in Appendix A.2 shows that the average difference between bids in the two types of auction is \$1.58, and this difference is statistically significant at the 1% level.

Result 3. In one-prize auctions, buyers overbid on average for almost all signals. In four-prize auctions, buyers overbid for low signals but underbid for high signals.

Result 4. In both types of auctions, average bidding functions show overbidding at low signals and less responsiveness to signals than predicted by the rational benchmark.

Evidence for results 3 and 4 can be seen in Figure 5, but we provide formal statistical analysis in Table 1.[26] Columns (1) and (2) estimate linear bidding functions in the one-prize and four-prize auctions, respectively. As discussed in Section 2.2, according to the symmetric Bayes Nash equilibrium the slopes of both functions should be equal to 3.5, and the intercepts should be 0 and 7.5, respectively.[27]

[24] As with the "Weak Winner's Curse" (see Footnote 23) we could define the "Weak Loser's Curse" as the empirical deviation payoffs being higher than the deviation payoffs implied by the Nash Equilibrium, which can be calculated at 29.166. While the empirical deviation payoffs are higher than this at 29.392, the difference is not significant ($p = 0.32$).

[25] It is important to highlight that both curses are equally detectable. Indeed, given our parameterization and under symmetric Nash (and symmetric χ-Cursed Equilibrium with common χ, as discussed in Section 4.2), the loser's deviation payoff in the four-prize auction has exactly the same distribution as the winner's payoff in the one-prize auction multiplied by -1. Empirically, the estimated standard error of average winner's payoff in the one-prize auction and average loser's deviation payoff in the four-prize auction are 0.25 and 0.23 respectively, implying comparable statistical precision for the estimated values.

[26] This table takes the exact form that we preregistered in our pre-analysis plan.

[27] We also estimated the pooled regression without fixed effects and reject the null hypotheses that average bids equal the Nash benchmark or the 'naïve/uncorrected' benchmark (i.e., treating opponents' bids as uninformative about their signals) with p-values less than 0.001; details are available upon request.

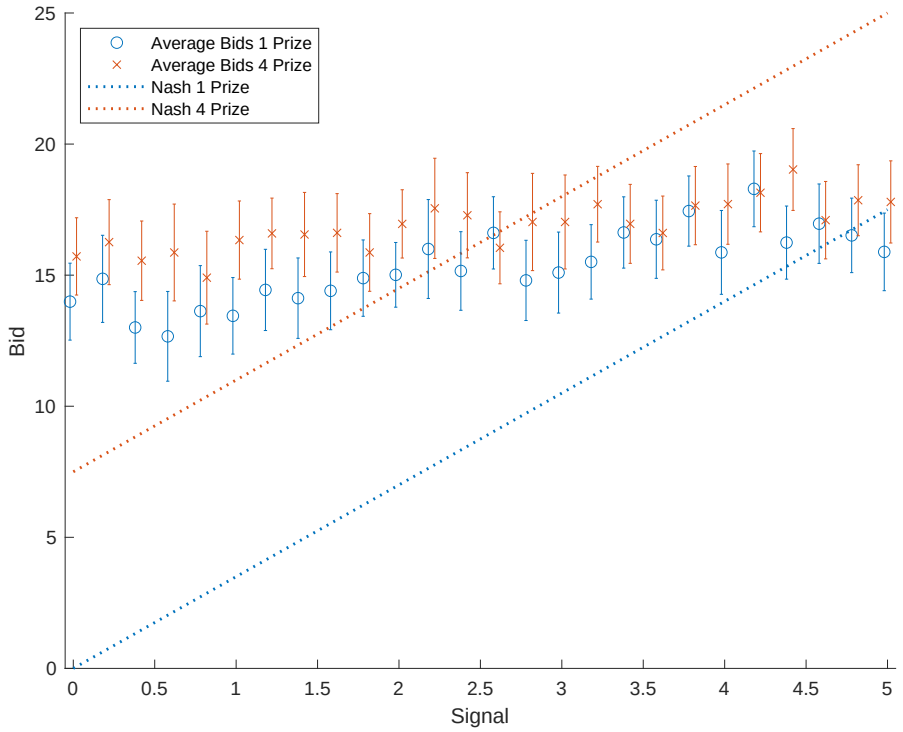
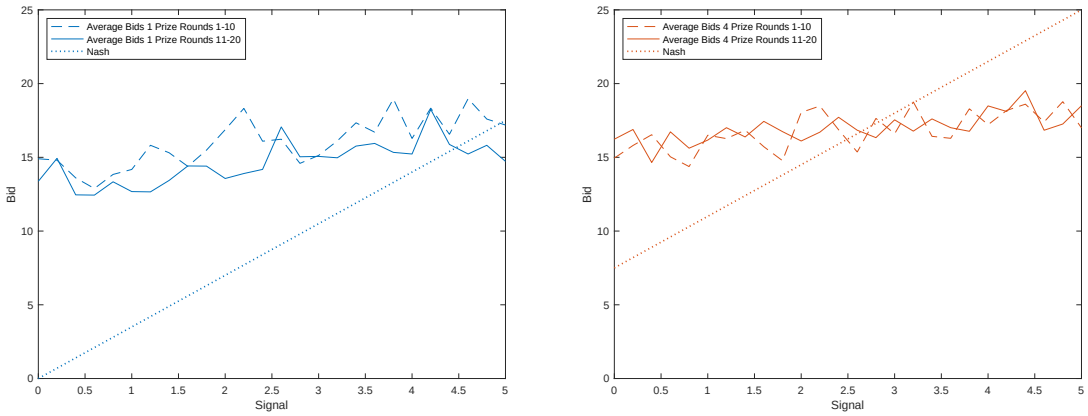


Figure 5: Averages vs. Theories in Auctions

Table 1: Choices

	(1) One-Prize-Bid	(2) Four-Prize-Bid	(3) Reservation Price
Signal	0.77***	0.65***	1.22***
	(0.091)	(0.077)	(0.17)
Highest Signal			-2.74***
			(0.76)
Second Highest Signal			-1.78**
			(0.69)
Third Highest Signal			-0.81
			(0.60)
Fourth Highest Signal			-0.67
			(0.54)
Constant	13.4***	15.3***	12.7***
	(0.22)	(0.19)	(0.34)
Observations	2400	2400	2400

Notes: Linear regression with subject- and round-fixed effects and standard errors clustered at the subject level. Significance indicated by: *** p<0.01, ** p<0.05, * p<0.1.



(a) One-Prize

(b) Four-Prize

Figure 6: Effects of Experience. There is slight learning in one-prize auctions, with bids decreasing significantly by rounds 11-20. Average bids in the four-prize auction do not change significantly with experience.

Figure 6 shows how average bidding functions change after subjects have become experienced, which we define as having completed ten rounds. From the figure, we see that once they have accumulated some experience with the task, subjects tend to lower their bids in the one-prize auction; however, there does not appear to be any effect of experience for four-prize auctions. One plausible reason for this asymmetry is that the Winner's Curse is often experienced as a realized monetary loss, whereas the Loser's Curse reflects a hypothetical, forgone gain, which may be less salient and therefore harder to detect and correct.[28]

The second column in Table 5 of Appendix A.2 shows that there is no statistically detectable difference between average bidding in the one-prize and four-prize auctions when subjects are inexperienced, but this gap increases to \$2.39 and becomes significantly different from zero once subjects have gained experience. Despite this, even with experience, it remains well below \$7.5, the level predicted by Bayes Nash equilibrium.

In Table 7 in Appendix A.4, we show how our results differ for subjects that have higher ability, as measured by above-median CRT scores or quiz scores. Because Table 1 already includes subject fixed effects, the intercept does not change. Instead, the table shows how responsiveness to the signal varies with ability level. The only relationship that is significant is the positive interaction between above-median CRT scores and the signal

[28]Prior research has shown that subjects often react more strongly to realized losses than hypothetical ones (Imas, 2016).

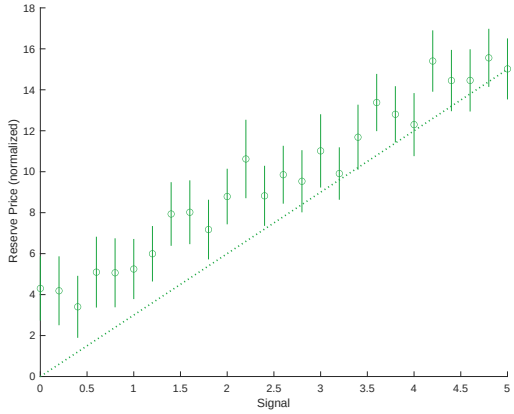


Figure 7: Average Value vs. Theory – Pooled and Rescaled

in one-prize auctions: subjects with higher CRT scores are roughly twice as responsive to their signal as subjects with lower CRT scores.

3.3 Failures of Contingent Thinking

We now turn to whether decision-making improves when subjects are directly provided with information that should be relevant for their decision. Figure 7 reports average willingness to pay in the non-strategic valuation task, conditional on signal and normalized across ranks by the theoretically predicted effect of the rank information (subtracting 10 conditional on the lowest rank, 7.5 conditional on the fourth-highest rank, etc.).[29] This figure provides evidence for Result 5.

Result 5. Average willingness to pay is slightly higher than the theoretically predicted rational benchmark for all signals.

Column (3) of Table 1 estimates a linear relationship between subjects' signals and their reservation prices, allowing for different intercepts by signal rank. It is instructive to compare the ratio of the estimated coefficients to their theoretical benchmarks across the auctions and the valuation task. In the one-prize and four-prize auctions, the estimated slopes of the bidding functions are 0.77 and 0.65, respectively, compared to the Nash equilibrium slope of 3.5. In contrast, the coefficient on signal in the valuation task is 1.22, relative to the risk-neutral benchmark of 3. This indicates that choices are more responsive to signals in the valuation task than in the auctions. The coefficients on the signal rank

[29] We report the average willingness to pay conditional on signal and split by rank in Online Appendix C.

dummies are correctly ordered (higher ranks are associated with lower valuations), but their magnitudes are substantially smaller than the rational benchmark.

We summarize how the differences between the observed choices and the rational benchmark vary across treatments in Table 2. In Column (1), the dependent variable is the difference between a subject's choice and the theoretical prediction, while the independent variables are indicators for the decision task being faced (with the valuation task being the omitted category). The results confirm substantial overvaluation in the valuation task and overbidding in the one-prize auctions ($p < 0.01$ for both tests). Overbidding in the four-prize auction is also statistically different from zero ($p = 0.04$), but the difference is substantially smaller. The dependent variable in column (2) is the *absolute value* of the difference between choice and theory. This provides the evidence for Result 6: The significant positive coefficients for both auctions indicate that average absolute differences between choices and the rational benchmark are smaller in the valuation task than in either type of auction.

Result 6. Decision-makers choose closer to the rational benchmark when provided with the relevant contingent information.

The valuation task is designed to remove the need for contingent reasoning. Because subjects are *given* rank information and no longer need to infer it from the auction's outcome, optimal behavior depends only on the subject's ability to form conditional expectations about V given (s_i, r).[30] Thus, while the missing Loser's Curse speaks to whether belief-based explanations can account for behavior across formats, evidence that choices move closer to the rational benchmark when rank information is made explicit speaks to failures of contingent reasoning. At the same time, the valuation task removes strategic interaction. We therefore treat the evidence from this task as suggestive about failures of contingent thinking and the formation of conditional expectations, while relying on cross-format auction comparisons to assess the explanatory scope of belief-based accounts.

This distinction also helps clarify why evidence consistent with failures of contingent thinking does not, by itself, imply the presence of a Loser's Curse. In models that operate *solely* through under-conditioning on the pivotal contingency, the same reasoning that generates a Winner's Curse when $K = 1$ typically predicts a Loser's Curse when $K = 4$ and the pivotal event flips from winning to losing. We do not find such a Loser's Curse in the

[30] Cursed equilibrium, by contrast, is an equilibrium concept: it combines optimization with a specific misperception of the information content of others' actions.

Table 2: Deviations from the Rational Benchmark

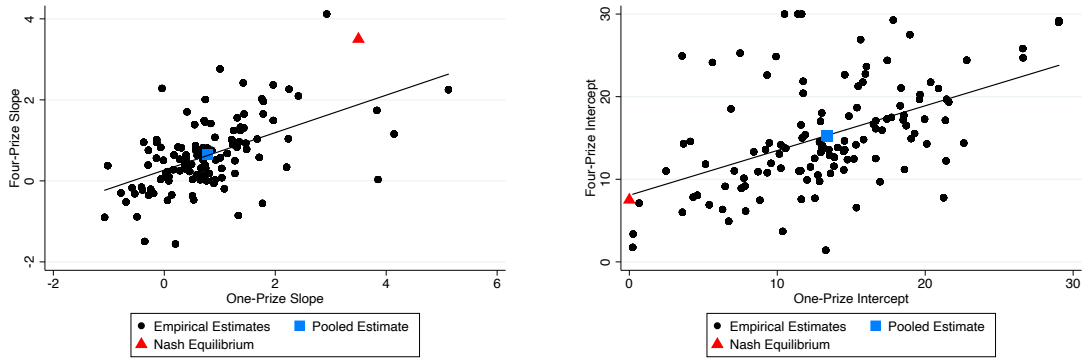
	(1) Diff. Theory	(2) Abs. Diff Theory
One-Prize	4.51***	2.62***
	(0.49)	(0.29)
Four-Prize	-1.41**	1.13***
	(0.57)	(0.28)
Constant	2.13***	5.91***
	(0.30)	(0.14)
Observations	7200	7200

Notes: Linear regression with subject-round fixed effects and standard errors clustered at the subject level. Significance indicated by: *** p<0.01, ** p<0.05, * p<0.1.

aggregate, suggesting that any under-conditioning force is not sufficient to account for the cross-format pattern on its own and may be offset by other forces that push bids upward.

Finally, behavior is closer to the rational benchmark in the valuation task than in auctions, but subjects still overbid in this non-strategic environment where beliefs about others are irrelevant. This pattern is consistent with an additional, more general upward-bidding force, which we examine in the next section.

In Table 6 of Appendix A.3, we report the second part of our preregistered analysis. Here, we analyze how decisions vary across the three tasks conditional on rank, and using subject-round fixed effects. Following the discussion in Section 2.2, we should observe that bids in the one-prize auction are higher than valuations for signal rank one, but lower than valuations for signal ranks two to five. Similarly, bids in the four-prize auction should be higher than valuations for signal ranks one to four, but lower than valuations for signal rank five. Instead, we find that bids are higher than valuations for both auctions and all signal ranks, although this relationship is not significant for ranks three to five in the one-prize auction. Furthermore, we show how the relationship between valuation and signal varies by subject ability in Table 8 of Appendix A.4. Subjects that have above-median CRT or quiz scores tend to be more responsive to both the magnitude and rank of their signal.



(a) Estimated Slopes

(b) Estimated Intercepts

Figure 8: Estimated Linear Bidding Functions. The line provides the best-fitting linear relationship between the estimated parameters.

4 Individual-Level Results: Comparison of Behavioral Models

In this section, we focus on behavior at the individual level, with the goal of identifying which behavioral model best describes each subject. Before considering the models, we summarize individual behavior in the auctions in Figures 8 and 9.

Figure 8 reports the coefficients resulting from estimates of linear bidding functions for each subject and relates them across auctions. Panel (a) shows the estimated slopes and panel (b) reports the estimated intercepts. Two conclusions can be drawn from these figures. First, most estimates are far from the rational (Nash Equilibrium) benchmark. Second, there is substantial heterogeneity *and consistency* at the individual level: Estimated parameters vary widely across individuals, and there is a strong positive relationship between estimates in the two auctions. These points suggest that behavioral models can be useful to explain behavior in these common-value auctions, and that there may be substantial heterogeneity in which model best captures an individual's behavior.

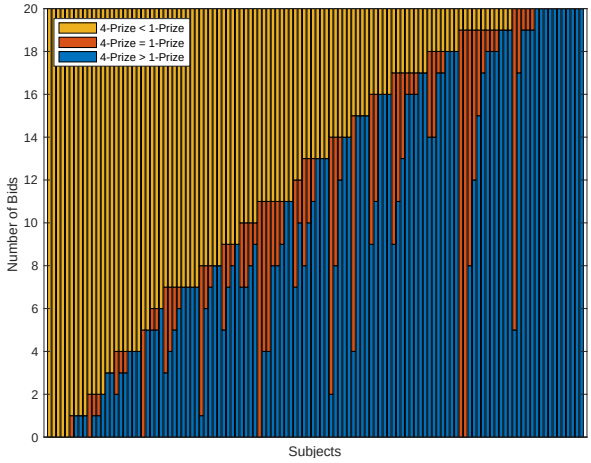


Figure 9: Bid Orders - Individuals

Additional evidence of the importance of heterogeneity can be found in Figure 9. For each subject, the figure reports the number of rounds in which their bid in the one-prize auction was strictly greater than, equal to, or strictly less than their bid in the four-prize auction. Eleven subjects always bid strictly higher in the four-prize auction than in the one-prize auction, while five subjects always bid strictly higher in the one-prize auction; recall that according to the rational benchmark, bids in the four-prize auction should always be higher than those in the one-prize auction. Moreover, as the figure shows, just a few subjects are responsible for the majority of rounds in which bids in the two auctions were exactly the same.

4.1 Individual-Level Classification

In the following sections, we report individual estimates for the four behavioral models we consider: Cursed Equilibrium, Level-k Thinking, Quantal Response Equilibrium, and Joy of Winning. To estimate the models, we assume that buyers choose according to a logit stochastic choice function given the model's parameters and the payoffs implied by those parameters. This allows us to estimate each model's parameters using Maximum Likelihood, while allowing for noisy decision-making.

Estimating the parameters of each model for each individual allows us to *classify* individuals according to the model that predicts their behavior the best. The models that we use differ in terms of the numbers of parameters that must be estimated for each of them, so we compare these models using the Akaike information criterion (AIC). Each subject is

classified into a behavioral model based on the model that minimizes AIC for their combined bidding data across both auction tasks.[31] The full set of AICs for each subject and model we consider can be found in Online Appendix D.

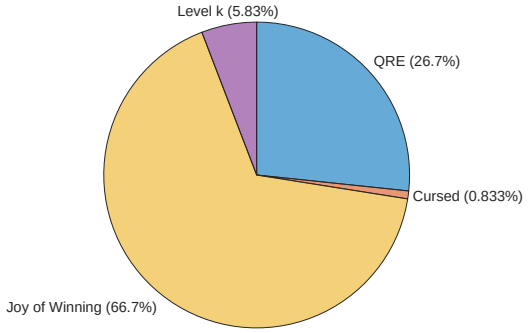


Figure 10: Proportions of Estimated Types

Figure 10 reports the proportions classified as each type. We can see that 80 subjects are classified as being best described by having a Joy of Winning. This is followed by Quantal Response Equilibrium and Level-k Thinking, with 32 and 7 subjects classified, respectively. Finally, 1 subject is classified as choosing according to Cursed Equilibrium.

4.2 Cursed Equilibrium

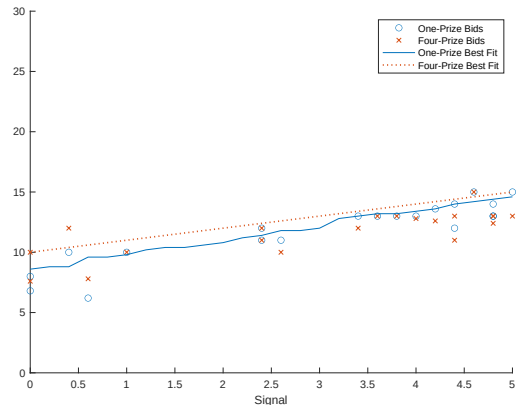
Motivated by the Winner's Curse in common-value auctions, Eyster and Rabin (2005) propose Cursed Equilibrium as an equilibrium concept. The essence of Cursed Equilibrium is that players correctly interpret their own private information and understand the average distribution of other players' actions, but fail to condition on the information that is implied by others' behavior.[32] More specifically, a cursed player incorrectly assigns probability $\chi \in [0, 1]$ to the other players playing their average distribution of actions irrespective of their type rather than their true, type-contingent strategy, to which she assigns probability $1 - \chi$; the parameter χ captures the extent of the bias, with $\chi = 0$ corresponding to

[31] We classify subjects using only their bidding data because the valuation task is nonstrategic, and all four models yield the same choice probabilities in that task under the logit stochastic choice assumption. Including valuation data would therefore not help discriminate between models.

[32] "Analogy-Based Expectations Equilibrium" (ABEE; Jehiel, 2005) and "Behavioral Equilibrium" (BE; Esponda, 2008) are equilibrium concepts closely related to cursedness. A fully cursed equilibrium coincides with the coarsest version of ABEE. In our setting, BE predicts *underbidding* for all values of K, and thus cannot explain the overbidding we see empirically.

χ_1 \ χ_4	≤ 0.05	$(0.05, 0.95)$	≥ 0.95
≤ 0.05	7 (0)	0 (0)	7 (0)
$(0.05, 0.95)$	1 (0)	0 (0)	2 (1)
≥ 0.95	23 (0)	4 (0)	37 (0)

(a) Estimated χ's



(b) Subject 96

Figure 11: Panel (a) classifies estimated values of χ_1 and χ_4 into bins, omitting 39 subjects with estimates of at least one of their logit parameters equal to zero. In parentheses, we give the number of subjects whose overall classification is cursed who fall into that bin. Panel (b) shows the bids and estimated modal choice of subject 96, whose overall classification was cursed and who had estimated parameters $\chi_1 = 0.408$ and $\chi_4 = 1$.

the fully rational, Bayesian benchmark, and $\chi = 1$ capturing the fully cursed case where a player assumes no connection between other players' actions and their types.

In our setting, the bidding functions of a cursed player take the following form

$$\begin{aligned} \beta_1^{\chi}(s_i) &= (1-\chi)\,\beta_1^*(s_i) + \chi\mathbb{E}[V|s_i] \\ &= 10\chi + (3.5 - 2.5\chi)\,s_i \end{aligned} \tag{7}$$

for $K = 1$, and

$$\begin{aligned} \beta_4^{\chi}(s_i) &= (1-\chi)\,\beta_4^*(s_i) + \chi\mathbb{E}[V|s_i] \\ &= 7.5 + 2.5\chi + (3.5 - 2.5\chi)\,s_i \end{aligned} \tag{8}$$

for $K = 4$.[33] It is worth highlighting that in the fully cursed case, expressions (7) and (8) both reduce to $s_i + 10$; hence a fully cursed buyer would bid the same in both auctions.

We assume that subjects will choose according to the logit stochastic choice function with noise parameter λ. Thus, we estimate values of χ and λ for each type of auction.[34]

[33] Notice that the convex-combination form is not a generic property of Cursed Equilibrium for arbitrary games. In our setting, it holds because the cursed belief replaces the type–action correlation with a mixture that preserves the marginal action distribution, implying a constant posterior mixing weight χ, and because the common value is linear in the signals; see Online Appendix B.1 for more details.

[34] Our estimation procedure is different from that used by Eyster and Rabin (2005). It estimates the cursed-

First, we compute a buyer's expected payoff in the experiment conditional on any combination of signal and bid for each auction.[35] We then calculate the "cursed" payoffs as those arising from having the correct beliefs about the likelihood of winning, but believing that the value of the prize will always be the signal plus 10. The estimated value of χ for an auction, then, is the weight on the cursed payoffs (with the remaining weight $1 - \chi$ on the true payoffs) that best fits the subjects' choices, given logit errors. We estimate the parameters restricting the values of χ_1 and χ_4 to be between 0 and 1 (as is generally assumed in models of partial cursedness) and λ to be weakly greater than zero (so choices are positively correlated with the decision-maker's perceived payoffs).

Panel (a) of Figure 11 shows the results of this estimation procedure.[36] The vast majority of subjects have estimated values of both χ_1 and χ_4 equal to either zero or one. The modal estimate is for the subject to be fully cursed in both auctions. Panel (b) of the same figure shows the decisions and estimated modal choices of Subject 96, who is the only subject who was classified as a cursed decision maker.

4.3 Level-k Thinking

Level-k Thinking is a structural non-equilibrium model introduced by Stahl and Wilson (1994, 1995) and Nagel (1995) that is meant to capture initial responses to games. Players in these games are assumed to have different levels of reasoning. A "level 0" type doesn't think strategically, choosing an action that seems natural or choosing at random. A "level 1" type assumes all other players are level 0 types and best responds to their beliefs. A "level 2" type assumes all other players are "level 1" and best responds, etc. The level-k model was used in Crawford and Iriberri (2007) to explain overbidding behavior in both private-value and common-value auctions.

To determine behavior predicted by the level-k model, the researcher must make an assumption about the precise behavior of the level 0 type. We follow Crawford and Iriberri

ness parameter based on comparing the best-response function implied by partial cursedness to the empirical distribution of bids. The estimated value of χ is the one that minimizes the sum of squared deviations between the two. Our procedure allows for a more direct comparison between cursedness and models that are inherently stochastic or that have set-valued best-response functions (i.e. QRE and Level-k).

[35] We compute this based on the joint empirical distribution of signals and bids. We draw four observations of bid-signal pairs at random from the full experiment. Given this artificial set of other bids, we compute the payoff a buyer would receive for every combination of bid and signal. We repeat this exercise a total of 100,000 times, and take the average across all repetitions to generate the payoffs that a buyer could expect for each combination of bid and signal.

[36] The figure omits 39 subjects for whom either χ_1 or χ_4 is not identified due to the estimated values of λ_1 or λ_4 being zero.

Types	1-Prize	4-Prize
R0	$\mathrm{U}[0, 30]$	$\mathrm{U}[0, 30]$
R1 & T0	$s_i + 10$	$s_i + 10$
R2 & T1	$\begin{cases} 10 \text{ if } s_i \leq 4 \\ 15 \text{ if} s_i > 4 \end{cases}$	$\begin{cases} 10 \text{ if } s_i \leq 1 \\ 15 \text{ if} s_i > 1 \end{cases}$
T2	$\begin{cases} < 10 \text{ if } s < 2 \\ [10, 15] \text{ if } 2 \leq s \leq \frac{1333}{369} \\ > 15 \text{ if } \frac{1333}{369} < s \end{cases}$	$\begin{cases} < 10 \text{ if } s < \frac{512}{369} \\ [10, 15] \text{ if } \frac{512}{369} \leq s \leq 3 \\ > 15 \text{ if } 3 < s \end{cases}$

Table 3: Best responses in the level-k model

(2007) who argue that this basic behavior may take one of two forms: a *random* level zero (R0) randomly chooses over the feasible range of bids uniformly without any relation to their private information, while a *truthful* level zero (T0) bids the value that their private information (but not the information conveyed by other buyers' behavior) would imply. We also follow Crawford and Iriberri (2007) in confining our attention to levels 0, 1, and 2, because prior evidence has shown higher levels to be comparatively rare. We summarize the best response behavior for the R0-R2 and T0-T2 types in Table 3.[37,38]

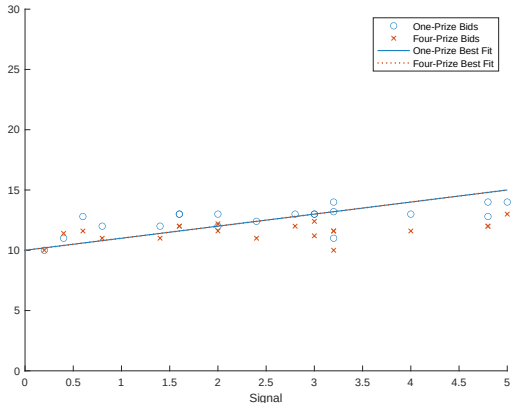
For each subject, we estimate three sets of logit parameters, each corresponding to the payoffs implied by the beliefs of one of the following types: (1) R1 or T0, (2) R2 or T1, or (3) T2. We then classify subjects into the category that has the highest log-likelihood. For all subjects that have estimated logit parameters equal to zero for all three types, we classify them as R0. The results of this process can be found in Panel (a) of Figure 12. Almost half of the subjects are classified as being R1 or T0 in both types of auctions, implying that their choices are best explained by beliefs that all other buyers are randomizing uniformly over all possible bids. Furthermore, all 7 of the subjects whose overall classification was as a level-k decision maker fall into this category.

[37]For the derivation of the best responses, see Online Appendix B.2.

[38]The best responses of R1 and fully cursed players coincide. However, the fact that the models differ in their assumptions about beliefs regarding others' behavior affects expected payoffs. These differences allow the models to be distinguished under the additional assumption of logit errors. Since beliefs are typically unspecified for T0 players, we adopt the convention that they hold the same beliefs as R1 players when computing their choice probabilities.

1-Prize \ 4-Prize	R0	R1 & T0	R2 & T1	T2
R0	0 (0)	6 (0)	14 (0)	1 (0)
R1 & T0	3 (0)	51 (7)	17 (0)	2 (0)
R2 & T1	2 (0)	7 (0)	2 (0)	1 (0)
T2	1 (0)	8 (0)	4 (0)	1 (0)

(a) Estimated Types



(b) Subject 109

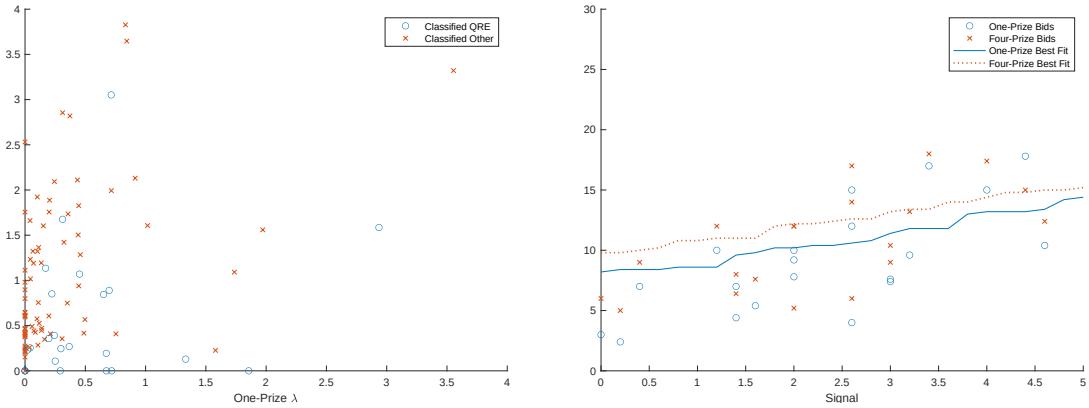
Figure 12: Panel (a) shows the number of subjects classified as each type within each auction. In parentheses, we give the number of subjects whose overall classification is level-k who fall into each type. Panel (b) shows the bids and estimated modal choices of subject 109, whose overall classification was level-k and whose subtypes were R1 in both auctions.

4.4 Quantal Response Equilibrium

Quantal Response Equilibrium (QRE) is a solution concept in which players act as if they have correct beliefs about the information and actions of others, but do not necessarily best respond to these beliefs (McKelvey and Palfrey, 1995). Instead, the likelihood that a choice is made is positively related to the payoffs that the choice leads to. Following much of the previous literature, we focus on the logit implementation of QRE, in which choice probabilities can be calculated based on a logit distribution (Goeree et al., 2016).

QRE has been successfully used to explain behavior in experimental auctions (Goeree et al., 2002; Camerer et al., 2016). In many types of common-value auctions, because players do not always choose in a deterministic and increasing way in relation to their signals, the inference that can be drawn from a set of bids is weaker and best-response functions are less steep than under Nash Equilibrium. We show computed QRE average bidding functions for various levels of the logit noise parameter in Online Appendix B.3.

In our empirical approach, we allow for subjects to have a different logit noise parameter; thus, our results are closer to the *heterogeneous* quantal response equilibrium discussed in Rogers et al. (2009). The relevant parameters that we estimate for QRE are the λ's from the logit stochastic choice function. We estimate these following the general approach discussed in Camerer et al. (2016). Specifically, we employ the same procedure that we use for



(a) Estimated λ's

(b) Subject 38

Figure 13: Panel (a) shows the estimated values of λ_1 and λ_4, omitting 19 subjects with extreme values of one of the parameters. Panel (b) shows the bids and estimated modal choice of subject 38, who had estimated parameters of $\lambda_1 = 2.94$ and $\lambda_4 = 1.59$.

Cursed Equilibrium (described in Section 3.1) to calculate the "true" payoffs that a buyer should expect to receive for any given bid and signal. Given these payoffs and the choices made by the subject, we estimate the logit parameters, restricting them to be greater than zero (so choices are required to be positively correlated with payoffs).

Panel (a) of Figure 13 shows the results of this estimation procedure. The estimated parameters have a correlation coefficient of 0.47 (statistically significantly different from zero with a p-value of less than 0.01). The estimates for subjects that are classified as QRE bidders are shown in blue circles, while those classified as choosing according to one of the other models have red x's. Those that are classified as QRE tend to have higher values of λ_1, the parameter that captures how noisy decision-making is in one-prize auctions. Panel (b) of the same figure shows the actual choices of a subject classified as QRE, along with the model's predicted modal choice.

4.5 Joy of Winning

As our last behavioral model, we consider the Joy of Winning model, which is the one that best describes the majority of our subjects. This model captures the idea that in addition to any monetary payoffs, bidders may directly care about winning the auction. Cox et al. (1988) and Cooper and Fang (2008) found that allowing for such a direct utility of winning was important to explain data from laboratory experiments on private-value auctions. To

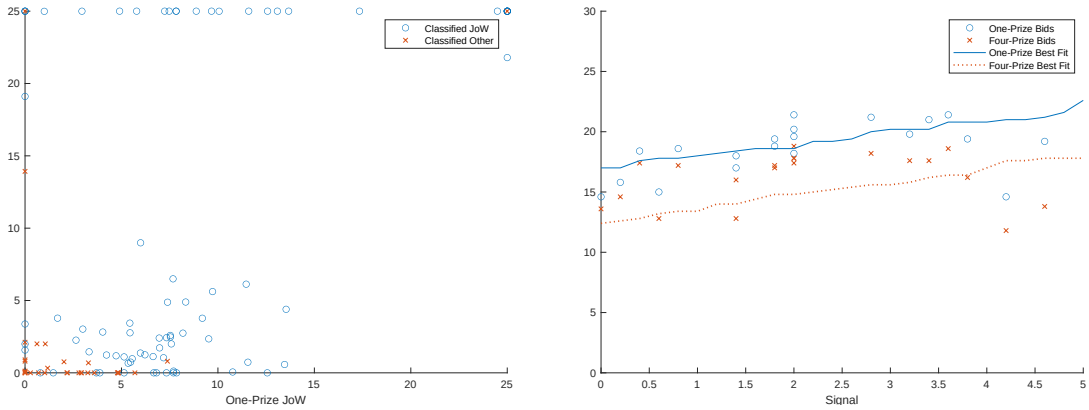
the best of our knowledge, however, with the exception of Van den Bos et al. (2008), prior work has not explicitly examined the Joy of Winning as a potential driver of overbidding and the Winner's Curse in common-value auctions.[39] We emphasize that we view Joy of Winning as a tractable stand-in for a broad family of mechanisms that lead to overbidding across auction settings, including non-monetary utility from winning, misperceived payoff structures, or "bidding fever" (Georganas et al., 2017; Gagnon-Bartsch et al., 2021; Nagel et al., 2025).

To account for the possibility that subjects may have different feelings about one-prize versus four-prize auctions, we allow the joy gained from winning to vary with each type of auction. We denote the monetary equivalent of the utility from winning a one-prize auction as w_1 and from winning a four-prize auction as w_4. We restrict both joy of winning parameters to be more than zero (so subjects must prefer winning) and less than 25 (the maximum monetary value of winning an auction).[40] We assume that the total utility a subject derives from placing a bid in the one-prize auction is based on the sum of two factors: the probability of winning with that bid multiplied by w_1 and the empirical average of monetary payoffs conditional on making that bid. The total utility from bids in the four-prize auction is calculated similarly. Finally, we assume that subjects make their choices based on this total utility using a logit stochastic choice model, which includes distinct noise parameters for each type of auction. As a result, we estimate two parameters related to the joy of winning and two logit noise parameters.

Panel (a) of Figure 14 shows the results of this estimation procedure. The correlation of w_1 and w_4 among the 110 subjects for whom both are identified is 0.51 (significantly different from 0 with a p-value of less than 0.01). The parameter estimates for subjects classified as behaving according to the Joy of Winning model are presented in blue. These parameters tend to be in one of two clusters. The first cluster has $w_4 = 25$—these are subjects that bid the maximum value in most or all of the four-prize auctions. The second cluster has positive and intermediate values of both w_1 and w_4, with the former being greater than the latter; this is consistent with buyers liking to win both auctions, but having a stronger preference to win when they would be the *unique* winner.

[39] Van den Bos et al. (2008) compares bidding behavior across three conditions: (1) subjects bidding against other human participants, (2) subjects bidding against computers emulating human bids, and (3) subjects bidding against computers playing the risk-neutral Nash equilibrium strategy. The study finds that overbidding decreases substantially when participants compete against computers, suggesting that winning against humans carries intrinsic value.

[40] Relaxing the restriction from 25 to 100 leads only one more subject to be classified as behaving according to the Joy of Winning model.



(a) Estimated Types

(b) Subject 56

Figure 14: Panel (a) shows the estimated joy of winning parameters, omitting 10 subjects with estimates of their logit parameters equal to zero. Panel (b) shows the bids and estimated modal choices of subject 56, who was classified as a Joy of Winning decision maker and whose joy of winning parameters were 7.3 in the one-prize auction and 2.4 in the four-prize auction.

4.6 Robustness of Classifications

We end this section by assessing the robustness of our classification exercise to alternative assumptions and estimation procedures.

First, we address our decision to allow parameter values to differ across the two auction types. This approach reflects the possibility that the auctions imposed different levels of complexity or cognitive demands. However, if the parameters reflect portable "deep" preferences or reasoning styles, they should be consistent across games. It is also possible that constraining parameters to be equal affects how subjects are classified. In practice, however, we find that classifications are largely stable: with the constraint imposed, 2 subjects are classified as following Cursed Equilibrium, 12 as Level-k Thinking, 33 as Quantal Response Equilibrium, and 73 as Joy of Winning (compared to 1, 7, 32, and 80, respectively, in Section 4.1).[41] Only 19 of 120 subjects change classification under the equal-parameters constraint. Thus, the conclusion that Joy of Winning best describes the majority of subjects holds even when parameters are restricted to be constant across auctions.

Second, we consider whether the estimation procedure might inherently favor the Joy of Winning model, potentially leading to misclassification even when other models better explain a subject's choices. To assess this, we generate an artificial dataset of 12,000

[41] The AICs for this procedure can also be found in Online Appendix D.

subjects who randomize uniformly over all available bids. We then apply the same classification procedure described in Section 4.1. Under this procedure, none of the artificial subjects are classified as cursed, 0.5% as Level-k, 92.6% as QRE, and 6.9% as having a Joy of Winning.[42] These results indicate that the estimation procedure does not systematically favor the Joy of Winning model.

5 Conclusion

There is extensive experimental evidence that people systematically overbid in common-value auctions, with subjects often falling prey to the Winner's Curse. Several behavioral models have been proposed to rationalize this finding, such as Cursed Equilibrium (Eyster and Rabin, 2005) and Level-k Thinking (Crawford and Iriberri, 2007). A perhaps less emphasized implication of these models is that when the pivotal contingency changes, the direction of the mistake should change as well: in common-value auctions with multiple items, bidders who under-react to the information conveyed by being pivotal should also be exposed to a Loser's Curse.

Our experiment was designed to provide a clean test of this cross-format implication by holding fixed the primitives of the environment (the signal structure, the common-value mapping, and the number of bidders) while varying the number of prizes. We find robust evidence of the Winner's Curse in the one-prize auction but no evidence of the Loser's Curse in the four-prize auction. We refer to this asymmetry as the missing Loser's Curse. We also find evidence suggesting that subjects struggle with contingent thinking: when the relevant contingency is made salient in a non-strategic valuation task by revealing signal rank, choices move closer to the rational benchmark, though substantial deviations remain. Overall, our results point to (i) a systematic tendency to bid high, consistent with a reduced-form "Joy of Winning" component, and (ii) significant noise and optimization errors, consistent with QRE. Importantly, we interpret Joy of Winning and QRE as parsimonious reduced-form representations of broader forces (such as a general propensity to overbid, "bidding fever," or difficulties forming conditional expectations), rather than as literal descriptions of preferences or cognition. We conclude by discussing some limitations of our study and directions for future research.

The missing Loser's Curse is informative for evaluating belief-based and contingent-

[42]This distribution is intuitive: uniform random choice is embedded in all four models when the logit parameters are zero, but QRE has the fewest parameters, so the AIC tends to favor it.

thinking explanations. In models such as Cursed Equilibrium or Level-k, the same under-conditioning that generates a Winner's Curse in a one-prize common-value auction typically also predicts a Loser's Curse once the pivotal event flips from winning to losing. Our evidence therefore suggests that mechanisms operating solely through under-conditioning on the pivotal contingency are unlikely to account for the full cross-format pattern of behavior in our setting; at the same time, we do not interpret the results as implying that belief-based explanations are irrelevant, but rather that additional forces are needed to match behavior across formats.

We did not find evidence of the Loser's Curse in our four-prize auctions. This is in contrast to Holt and Sherman (1994), who found evidence for the Loser's Curse under the experimental treatment where it was supposed to arise. However, a few differences in the experimental designs might contribute to explain these opposite findings. Holt and Sherman (1994) consider a bilateral-trade problem rather than an auction. In their study, buyers, who do not know their valuation but do know it is 1.5 times that of the seller, make a take-it-or-leave-it offer to a seller whose behavior is simulated by a computer.[43] Holt and Sherman (1994) interpret their simultaneous finding of both the Winner's Curse and the Loser's Curse (under different experimental conditions) as supporting a model of naïve bidding (which is equivalent to fully cursed equilibrium) over one with a Joy of Winning. In our auction experiment, instead, the results go in the opposite direction; yet, we believe that it is more natural for subjects to experience Joy of Winning when competing in an auction against other humans than when submitting offers to a computerized seller. The empirical differences between our studies suggest that behavior in one setting might not match behavior in the other. Linking behavior across the two settings is a promising avenue for future research.

While our experimental setting and design differ from those of Nagel et al. (2025), the two papers offer complementary insights into the sources of overbidding and the Winner's Curse in common-value auctions. Broadly, Nagel et al. (2025) attribute the Winner's Curse to (1) difficulties in computing order statistics, (2) failures of contingent thinking, and (3) errors stemming from "bidding fever" or general computational mistakes in payoff calculations. Our finding that subjects continue to "overbid" in the valuation task (where choices are non-strategic and do not require contingent reasoning) provides evidence that some overbidding arises from difficulty in calculating conditional expectations. We also empha-

[43]Notice that, using a design similar to Holt and Sherman (1994), Fudenberg and Peysakhovich (2016), like us, find no evidence of the Loser's Curse.

size the role of contingent reasoning: when relevant contingencies are made salient, the average absolute deviation from the theoretical benchmark declines. Finally, we attribute the pervasive overbidding in our auction treatments to the Joy of Winning model, which we interpret as a reduced-form representation of general overbidding forces including bidding fever and computational frictions.

For our classification exercise in Section 4, we used two distinct empirical approaches. For the purpose of estimating the parameters of QRE, Cursed Equilibrium, and Joy of Winning, we first used the empirical distribution of bids to compute buyers' expected monetary payoffs, and then used those payoffs to estimate the behavioral parameters. This contrasts with the approach we used for Level-k thinking, where we made specific assumptions about what buyers of different levels believe their payoffs would be, and then estimated the levels based on these beliefs.[44] It is possible that the estimated parameters might be different if the empirical distribution of payoffs played a role in subjects' hypothesized beliefs.

Finally, while behavior in our study was closer to the rational benchmark in the valuation task than in the auctions, subjects' choices in the former setting still displayed a clear tendency to overbid. Indeed, despite the effects of rank information found in Table 1 following the correct order (the effect for the highest signal is the most negative, followed by the second-highest signal, etc.), they are not fully distinguishable statistically, and are quite far from the rational prediction. Moreover, the estimated responsiveness of valuations to the signal conditional on rank information is not statistically distinguishable from 1, and thus much lower than the rational benchmark of 3 (cf. expression (6)). We believe that subjects' inability to correctly compute conditional expectations is likely to have an important effect across many economic settings, as noted also by Ngangoué and Schotter (2023) and Nagel et al. (2025); hence, an important direction for future work is to measure these limitations more directly (e.g., via incentivized expectation elicitation) and to study how they interact with competitive motives and feedback.

[44]Camerer et al. (2016) use the same approach of relying on empirical payoff distributions for QRE and Cursed Equilibrium while using the theoretical payoffs for Level-k. Goeree et al. (2016) refer to the use of the empirical distribution of bids as the "empirical payoff approach", as compared to the "equilibrium correspondence approach" of relying solely on theoretical assumptions about payoffs.

A Appendix

In this appendix, we provide additional analysis and results omitted from the main text.

A.1 Demographics

We begin by reporting our sample's demographics and summary statistics.

Table 4: Summary Statistics

	Mean	Std. Dev.
CRT Score	1.53	1.18
Female	0.59	0.49
Age	23.43	4.79
English	0.23	0.42
Economics	0.33	0.47
Subjects	120.00	

Notes: CRT Score is the number of correct answers on a Cognitive Reflection Test, ranging from 0 to 3. Female, English, and Economics are equal to one if the subjects report being female, speaking English as a first language, and majoring in Economics, respectively.

A.2 Experience

In this section, we report regression analysis of how subject's choices vary with experience. Table 5 shows the results. Column (1) demonstrates that bids in the Four-Prize auction are statistically significantly higher than those in the One-Prize auction. The second column demonstrates that this result arises primarily as subjects gain experience: there is no statistically significant difference in the first 10 rounds, but the difference becomes apparent in the second 10 rounds.

Table 5: Choices

	(1) Choice	(2) Choice
Four-Prize	1.58***	0.77
	(0.55)	(0.56)
Reservation	-0.74	-1.25**
	(0.50)	(0.49)
Experienced × Four-Prize		1.62***
		(0.38)
Experienced × Reservation		1.02**
		(0.48)
Constant	15.3***	15.3***
	(0.29)	(0.29)
Observations	7200	7200

Notes: Linear regression with subject-round fixed effects and standard errors clustered at the subject level. "Experienced" is a dummy that is equal to one for bids made in rounds 11-20. Significance indicated by: *** p<0.01, ** p<0.05, * p<0.1.

A.3 Signal Rank

Next, we analyze how subjects' decisions vary across tasks conditional on signal rank. Columns (1)-(5) restrict to data only from subjects who are told they have the corresponding signal rank. If the bounds on bidding discussed in Section 2.2 held in the data, the coefficients on "One-Prize" would be negative for all but the highest signal, and the coefficients on "Four-Prize" would be positive for all but the lowest signal. Instead, all coefficients are positive (with many of them statistically significantly so). This is consistent with the general observation of systematic overbidding in the auctions, in excess of what is observed in the purchase stage.

Table 6: Choices by Signal Rank

	(1) Choice	(2) Choice	(3) Choice	(4) Choice	(5) Choice	(6) Choice
One-Prize (Highest Signal)	1.60***					1.60***
	(0.56)					(0.56)
Four-Prize (Highest Signal)	3.09***					3.09***
	(0.69)					(0.69)
One-Prize (2nd Highest Signal)		1.34**				1.34**
		(0.60)				(0.60)
Four-Prize (2nd Highest Signal)		2.37***				2.37***
		(0.70)				(0.70)
One-Prize (3rd Highest Signal)			0.15			0.15
			(0.60)			(0.60)
Four-Prize (3rd Highest Signal)			2.19***			2.19***
			(0.66)			(0.66)
One-Prize (4th Highest Signal)				0.57		0.57
				(0.74)		(0.74)
Four-Prize (4th Highest Signal)				2.18***		2.18***
				(0.76)		(0.76)
One-Prize (Lowest Signal)					0.051	0.051
					(0.59)	(0.59)
Four-Prize (Lowest Signal)					1.76***	1.76***
					(0.57)	(0.57)
Constant	14.9***	14.7***	15.0***	14.1***	14.0***	14.5***
	(0.37)	(0.38)	(0.35)	(0.44)	(0.34)	(0.31)
Observations	1440	1440	1440	1440	1440	7200

Notes: Linear regression with subject-round fixed effects and standard errors clustered at the subject level. Significance indicated by: *** p<0.01, ** p<0.05, * p<0.1.

A.4 Subjects' Understanding

The next two tables display how subjects' behavior varies with ability in the auctions and in the valuation task, respectively.

Table 7: Bidding and Subject Understanding

	(1) 1-Prize-Bid	(2) 1-Prize-Bid	(3) 4-Prize-Bid	(4) 4-Prize-Bid
Signal	0.67***	0.52***	0.67***	0.58***
	(0.19)	(0.13)	(0.14)	(0.11)
High Quiz Score × Signal	0.15		-0.041	
	(0.21)		(0.17)	
High CRT × Signal		0.48***		0.13
		(0.18)		(0.16)
Constant	13.4***	13.4***	15.3***	15.3***
	(0.22)	(0.22)	(0.19)	(0.19)
Observations	2400	2400	2400	2400

Notes: Linear regression with subject- and round-fixed effects and standard errors clustered at the subject level. Significance indicated by: *** p<0.01, ** p<0.05, * p<0.1. "High Quiz Score" equals 1 for scores $\geq 6/7$; "High CRT" equals 1 for scores $\geq 2/3$.

Table 8: Valuations and Subject Understanding

	(1) Reservation Price	(2) Reservation Price	(3) Reservation Price
Signal	1.33***	1.65***	1.22***
	(0.20)	(0.25)	(0.17)
Highest Signal	-3.85***	-4.65***	-2.74***
	(0.84)	(0.88)	(0.76)
Second Highest Signal	-2.88***	-3.26***	-1.78**
	(0.74)	(0.85)	(0.69)
Third Highest Signal	-1.73***	-2.15***	-0.81
	(0.61)	(0.71)	(0.60)
Fourth Highest Signal	-1.52***	-1.76***	-0.67
	(0.54)	(0.56)	(0.54)
Constant	12.8***	12.2***	12.7***
	(0.32)	(0.29)	(0.34)
High Quiz Score	Yes	No	No
High CRT	No	Yes	No
Observations	1740	1280	2400

Notes: Linear regression with subject- and round-fixed effects and standard errors clustered at the subject level. Significance indicated by: *** p<0.01, ** p<0.05, * p<0.1. Column (1) includes subjects with quiz scores $\geq 6/7$; column (2) includes subjects with CRT scores $\geq 2/3$.

A.5 Order Effects

In this section, we consider whether the fixed ordering of decisions within each round may have influenced subjects' behavior. In particular, the purchase stage always followed the bidding stage,[45] and within the bidding stage, the slider for the four-prize auction was always positioned below the one-prize slider. Because this ordering was not randomized across subjects, it raises the possibility that experience with earlier tasks in a round influenced later decisions.

While possible, we consider this *a priori* unlikely. The two auction decisions were presented on the same screen, and subjects could make their bids in either order or revise them freely. Moreover, unlike in typical within-subject designs where subjects complete one treatment before moving to another, all three treatments were presented in each round. Any learning or spillovers would therefore apply across treatments. Consistent with this, Figure 6 shows only a minor effect of experience on average bids, and only in the one-prize auction.

To further assess possible order effects, we re-estimate our main within-round comparisons by shifting the timing of treatments across rounds. Specifically, in Tables 9 and 10, Column (1) replicates the original specification from Table 2. Column (2) re-labels one-prize bids as if they occurred one round later, effectively comparing those bids to the subsequent round's four-prize bids and valuations.[46] Columns (3) and (4) conduct analogous shifts for the four-prize and valuation tasks. The resulting changes in coefficients are negligible, indicating that ordering across tasks does not materially affect our estimates.

[45] This structure was necessary to preserve equilibrium conditions in the auction: providing rank feedback before bidding would have altered strategic incentives.

[46] The number of observations is lower in columns (2)-(4) because the regression drops the observations from the final round of the shifted treatment. These observations would be fully explained by the subject-round fixed effects.

Table 9: Order Effects in Deviations from the Rational Benchmark

	(1) Diff. Theory	(2) Diff. Theory	(3) Diff. Theory	(4) Diff. Theory
One-Prize	4.51***	4.48***	4.51***	4.57***
	(0.49)	(0.49)	(0.49)	(0.50)
Four-Prize	-1.41**	-1.41**	-1.47**	-1.35**
	(0.57)	(0.57)	(0.58)	(0.57)
Constant	2.13***	2.16***	2.14***	2.10***
	(0.30)	(0.30)	(0.30)	(0.31)
Shifted Treatment	-	One-Prize	Four-Prize	Val. Task
Observations	7200	7080	7080	7080

Notes: Linear regression with subject-round fixed effects and standard errors clustered at the subject level. Significance indicated by: *** p<0.01, ** p<0.05, * p<0.1.

Table 10: Order Effects in Absolute Deviations from the Rational Benchmark

	(1) AD Theory	(2) AD Theory	(3) AD Theory	(4) AD Theory
One-Prize	2.62***	2.73***	2.62***	2.60***
	(0.29)	(0.30)	(0.29)	(0.29)
Four-Prize	1.13***	1.13***	1.18***	1.11***
	(0.28)	(0.28)	(0.28)	(0.27)
Constant	5.91***	5.89***	5.90***	5.92***
	(0.14)	(0.14)	(0.14)	(0.14)
Shifted Treatment	-	One-Prize	Four-Prize	Val. Task
Observations	7200	7080	7080	7080

Notes: Linear regression with subject-round fixed effects and standard errors clustered at the subject level. "AD Theory" refers to the absolute difference between the choice and the rational benchmark. Significance indicated by: *** p<0.01, ** p<0.05, * p<0.1.

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Supplemental Material

This Online Appendix is organized as follows. Section B shows the derivation of strategies for some of the behavioral models discussed in the text. Section C reports the average choices in the purchase stage conditional on signal and rank. Section D reports the AICs for each of the models estimated and discussed in the main text. Section E provides screenshots from the experiment.

B Computation of Strategies Under Behavioral Models

B.1 Cursed Equilibrium

In this section, we derive the χ-cursed bidding functions stated in Section 4.2. Recall the environment from Section 2.1: there are $N \geq 2$ bidders and K identical items, with $N > K \geq 1$. Bidder i observes a signal (type) $s_i \in [\underline{s}, \bar{s}] \subset \mathbb{R}_+$; signals are i.i.d. with cdf F and density $f > 0$, and the common value is

$$V(s_1, \ldots, s_N) = \sum_{i=1}^{N} s_i. \tag{A.1}$$

In a uniform-price auction, the K highest bidders each obtain one item and pay the highest rejected bid (the $(K+1)$-st highest bid). A bidder's payoff is $V - p$ if she wins and 0 otherwise.

Fix a bidder i and let $y_1 > y_2 > \cdots > y_{N-1}$ denote the order statistics of her competitors' types. As stated in the main text, the unique symmetric Bayes–Nash equilibrium bidding function in the uniform-price auction for K items is

$$\beta_K^*(s_i) = \mathbb{E}[\, V \,|\, y_K = s_i,\ s_i\,]\,. \tag{A.2}$$

A χ-cursed bidder assigns probability $\chi \in [0, 1]$ to her opponents playing their *average distribution of actions* irrespective of their type, and probability $1 - \chi$ to opponents playing the true type-contingent strategy. Let $\mu(\cdot \mid y_K = s_i,\ s_i)$ denote the Bayesian conditional distribution of opponents' types and denote by $\mu(\cdot \mid s_i)$ the (unconditional) distribution of opponents' types given s_i (which coincides with the prior over opponents' types because

signals are independent). Under χ-cursedness, bidder i uses the *mixture* belief

$$\mu^{\chi}(\cdot \mid y_K = s_i,\ s_i) = (1-\chi)\,\mu(\cdot \mid y_K = s_i,\ s_i) + \chi\,\mu(\cdot \mid s_i). \tag{A.3}$$

The following lemma shows that a χ-cursed bidder partially fails to condition on the information implied by her opponents' behavior (here, the contingency $y_K = s_i$).

Lemma A.1 (Mixture identity)**.** *For any integrable function g of the opponents' types,*

$$\begin{aligned}\mathbb{E}^{\chi}[g(y_1,\ldots,y_{N-1}) \,|\, y_K = s_i,\ s_i] =& (1-\chi)\,\mathbb{E}[g(y_1,\ldots,y_{N-1}) \,|\, y_K = s_i,\ s_i] \\ &+ \chi\,\mathbb{E}[g(y_1,\ldots,y_{N-1}) \,|\, s_i]\,,\end{aligned} \tag{A.4}$$

where $\mathbb{E}^{\chi}[\cdot]$ denotes expectation taken under μ^{χ}.

Proof. This follows immediately from the definition of μ^{χ} in (A.3) and linearity of expectation:

$$\mathbb{E}^{\chi}[g \mid y_K = s_i, s_i] = \int g\, d\mu^{\chi} = (1-\chi)\int g\, d\mu + \chi \int g\, d\mu(\cdot \mid s_i).$$

□

Next, let $r(s_i) := \mathbb{E}[V \mid s_i]$, i.e., bidder i's expected value of the prize conditional on her signal only. The next proposition characterizes the χ-cursed equilibrium in uniform-price auctions.

Proposition A.1 (χ-cursed equilibrium bid in the uniform-price auction)**.** *In the uniform-price auction for K items, the symmetric χ-cursed equilibrium bidding function satisfies*

$$\beta_K^{\chi}(s_i) = \mathbb{E}^{\chi}[\,V \,|\, y_K = s_i,\ s_i] = (1-\chi)\,\beta_K^{*}(s_i) + \chi\,\mathbb{E}[V \mid s_i] = (1-\chi)\,\beta_K^{*}(s_i) + \chi\, r(s_i). \tag{A.5}$$

Proof. Fix bidder i with signal s_i. Consider a deviation in which she bids as if her type were t (equivalently, chooses the bid that a type-t bidder would choose under the candidate increasing symmetric strategy). Under monotonicity, bidder i wins if and only if $y_K \leq t$, and if she wins, the (uniform) price equals the highest rejected bid, which is the K-th highest competing bid and therefore a function of y_K. Standard arguments (as in the derivation of (A.2)) imply that an interior optimal bid satisfies the first-order condition that

the bidder is indifferent at the margin; i.e., at $y_K = t$ the bid equals the bidder's expected value conditional on being pivotal:

$$\beta_K^\chi(t) = \mathbb{E}^\chi[\, V \,|\, y_K = t,\ s_i]\,.$$

In a symmetric equilibrium, $t = s_i$, hence

$$\beta_K^\chi(s_i) = \mathbb{E}^\chi[\, V \,|\, y_K = s_i,\ s_i]\,.$$

Finally, apply Lemma A.1 with $g = \sum_{j\neq i} s_j$ and use $V = s_i + \sum_{j\neq i} s_j$ to obtain

$$\mathbb{E}^\chi[V \mid y_K = s_i, s_i] = (1-\chi)\mathbb{E}[V \mid y_K = s_i, s_i] + \chi\mathbb{E}[V \mid s_i] = (1-\chi)\beta_K^*(s_i) + \chi r(s_i),$$

which is (A.5). □

B.1.1 Closed forms for the experimental parameters ($N = 5$, $s_i \sim U[0,5]$)

In the experiment, $N = 5$ and $s_i \sim U[0,5]$. Hence $\mathbb{E}[s_j] = 2.5$ and

$$r(s_i) = \mathbb{E}[V \mid s_i] = s_i + 4 \cdot 2.5 = s_i + 10. \tag{A.6}$$

Case $K = 1$ (one-prize / second-price auction). Here $y_1 = \max\{s_j : j \neq i\}$. Conditional on $y_1 = y$, one opponent has signal y and the remaining three opponents' signals are i.i.d. $U[0,y]$, so their conditional mean is $y/2$. Therefore,

$$\mathbb{E}[V \mid y_1 = y,\ s_i] = s_i + y + 3 \cdot \frac{y}{2} = s_i + \frac{5}{2}y.$$

Imposing the pivotality condition $y_1 = s_i$ yields

$$\beta_1^*(s_i) = \mathbb{E}[V \mid y_1 = s_i,\ s_i] = s_i + \frac{5}{2}s_i = 3.5\, s_i. \tag{A.7}$$

Substituting (A.6) and (A.7) into (A.5) gives

$$\beta_1^\chi(s_i) = (1-\chi)\, 3.5 s_i + \chi(s_i + 10) = 10\chi + (3.5 - 2.5\chi)s_i, \tag{A.8}$$

which matches equation (7) in the main text.

Case $K = 4$ (four-prize / fifth-price auction). Here $y_4 = \min\{s_j : j \neq i\}$. Conditional on $y_4 = y$, one opponent has signal y and the remaining three opponents' signals are i.i.d. $U[y, 5]$, so their conditional mean is $(y + 5)/2$. Therefore,

$$\mathbb{E}[V \mid y_4 = y,\ s_i] = s_i + y + 3 \cdot \frac{y+5}{2} = s_i + \frac{5}{2}y + \frac{15}{2}.$$

Imposing the pivotality condition $y_4 = s_i$ yields

$$\beta_4^*(s_i) = \mathbb{E}[V \mid y_4 = s_i,\ s_i] = s_i + \frac{5}{2}s_i + \frac{15}{2} = 7.5 + 3.5\, s_i. \tag{A.9}$$

Substituting (A.6) and (A.9) into (A.5) gives

$$\beta_4^\chi(s_i) = (1-\chi)\,(7.5 + 3.5 s_i) + \chi(s_i + 10) = 7.5 + 2.5\chi + (3.5 - 2.5\chi)s_i, \tag{A.10}$$

which matches equation (8) in the main text.

B.2 Level-k

In this section, we derive the best responses of level-k types reported in Table 3, starting with the one-prize auction.

B.2.1 One-prize auction

A random level-0 (R0) player randomizes uniformly over the support of feasible bids:

$$\beta_1^{R0}(s_i) \sim U(0, 30). \tag{A.11}$$

A random level-1 (R1) player believes her opponents are R0 players who bid according to (A.11) and therefore knows that there is no inference that she could draw from their bids. Let $r(s_i) := \mathbb{E}[V|s_i]$ denote the expected value of the item for bidder i conditional on just her signal. Then, bidder i solves the following problem

$$\max_{b_i} \int_0^{b_i} (r(s_i) - x)\, f_1(x)\, dx$$

where $f_1(x)$ denotes the density of the highest competing bid. The FOC yields

$$\left(r\left(s_i\right)-b_i\right) f_1\left(b_i\right)=0 \iff b_i=r\left(s_i\right).$$

Hence, for the parameters of our experiment, the R1 bidding strategy is given by

$$\beta_1^{R1}\left(s_i\right)=10+s_i \tag{A.12}$$

which coincides with the fully cursed one.

A random level-2 (R2) player thinks her opponents are R1 players who bid according to (A.12); therefore, because this expression is increasing, she will adjust her common-value estimate for the information revealed by winning. Let $v\left(s_i, y\right):=\mathbb{E}\left[V \mid s_i, s_1=y\right]$ denote the expected value of the item for bidder i conditional on her signal and on the value of the signal of her highest competitor. Then, bidder i solves the following problem

$$\max_{b_i} \int_0^{\beta_1^{R1^{-1}}\left(b_i\right)}\left(v\left(s_i, y\right)-\beta_1^{R1}(y)\right) f_1(y)\, d y .$$

For the parameters of our experiment and when $b_i \in[10,15]$, the expression above becomes

$$\max_{b_i} \frac{1}{625}\left(b_i-10\right)^4\left(1.2 b_i+s_i-22.0\right).$$

Clearly, $b_i=10$ would ensure that bidder i loses the auction and attains a payoff of zero. Moreover, bidder i's expected payoff is negative for any bid between 10 and 15 as long as her signal is lower than 4, and otherwise it is increasing in b_i conditional on winning. Hence, for the parameters of our experiment, the R2 bidding strategy is given by

$$\beta_1^{R2}\left(s_i\right)=\left\{\begin{array}{lll} 10 & \text{if} & s_i \leq 4 \\ 15 & \text{if} & s_i>4 \end{array}\right. . \tag{A.13}$$

Thus, if $s_i \leq 4$, she will bid so low as to ensure she loses; if instead $s_i>4$, she will bid high enough to guarantee herself to win.

Next, consider a truthful level-0 (T0) player; this player bids the expected value of the prize conditional on just her signal:

$$\beta_1^{T0}\left(s_i\right)=10+s_i. \tag{A.14}$$

Notice that $\beta_1^{T0}\left(s_i\right)=\beta_1^{R1}\left(s_i\right)$.

A truthful level-1 (T1) player believes all of her opponents to be T0 players. In turn, her best response coincides with that of an R2 player; i.e., $\beta_1^{T1}(s_i) = \beta_1^{R2}(s_i)$.

A truthful level-2 (T2) player believes all of her opponents to be T1 players who bid according to β_1^{T1}. Given that T1 players bid either 10 or 15, a T2 player has three relevant options: i) bid strictly less than 10, guaranteeing herself a loss, ii) bid between 10 and 15, in which case she wins at a price of 10 but only if all of her opponents' signals are lower than 4, or iii) bid strictly more than 15, guaranteeing herself a win. Clearly, her expected payoff from bidding strictly less than 10 is equal to zero. Any bid between 10 and 15 yields the following expected payoff

$$\left(\frac{4}{5}\right)^4 [s_i + 4(2) - 10] \tag{A.15}$$

where $s_i + 4(2)$ denotes i's expectation of the value of the prize conditional on all of her opponents' signals being lower than 4. It is easy to see that expression (A.15) is strictly positive if and only if $s_i > 2$.

If she bids strictly more than 15, the bidder wins for sure and her expected payoff equals

$$\left(\frac{4}{5}\right)^4 [s_i + 4(2) - 10] + \left[1 - \left(\frac{4}{5}\right)^4\right] \left[s_i + \frac{4202}{369} - 15\right] \tag{A.16}$$

where $s_i + \frac{4202}{369}$ denotes i's expectation of the value of the prize conditional on at least one of her opponents' signals being higher than 4. It is easy to verify that (A.16) exceeds (A.15) if and only if $s_i > \frac{1333}{369}$.

Hence, for the parameters of our experiment, the L2 bidding strategy is given by

$$\beta_1^{R2}(s_i) = \begin{cases} < 10 & \text{if} \quad s_i \leq 2 \\ \in [10, 15] & \text{if} \quad 2 < s_i \leq \frac{1333}{369} \\ > 15 & \text{if} \quad \frac{1333}{369} < s_i \end{cases}. \tag{A.17}$$

B.2.2 Four-prize auction

Consider now the four-prize auction. A random level-0 (R0) player randomizes uniformly over the support of feasible bids:

$$\beta_4^{R0}(s_i) \sim U(0, 30). \tag{A.18}$$

A random level-1 (R1) player believes her opponents are R0 players who bid according to (A.18) and therefore knows that there is no inference that she could draw from the bids of her opponents. Then, bidder i solves the following problem

$$\max_{b_i} \int_0^{b_i} \left(r\left(s_i\right) - x\right) f_4\left(x\right) dx$$

where $f_4\left(x\right)$ denotes the density of the fourth-highest competing bid. The FOC yields

$$\left(r\left(s_i\right) - b_i\right) f_4\left(b_i\right) = 0 \iff b_i = r\left(s_i\right).$$

Hence, for the parameters of our experiment, the R1 bidding strategy is given by

$$\beta_4^{R1}\left(s_i\right) = 10 + s_i \tag{A.19}$$

which again coincides with the fully cursed one.

A random level-2 (R2) player believes her opponents are R1 players who bid according to (A.19); therefore, because this expression is increasing, she will adjust her common-value estimate for the information revealed by winning. Let $w\left(s_i, y\right) := \mathbb{E}\left[V | s_i, s_4 = y\right]$ denote the expected value of the item for bidder i conditional on her signal and on the value of the signal of her fourth-highest competitor. Then, bidder i solves the following problem

$$\max_{b_i} \int_0^{\beta_4^{R1^{-1}}(b_i)} \left(w\left(s_i, y\right) - \beta_4^{R1}\left(y\right)\right) f_4\left(y\right) dy.$$

For the parameters of our experiment and when $b_i \in [10, 15]$, the expression above becomes

$$\max_{b_i} \; - \frac{\left(b - 15\right)^4 \left(6b + 5s - 65\right)}{3125} + s - 1.$$

Clearly, $b_i = 10$ would ensure that bidder i loses the auction and attains a payoff of zero. Moreover, bidder i's expected payoff is negative for any bid between 10 and 15 as long as her signal is lower than 1, and otherwise it is increasing in b_i conditional on winning. Hence, for the parameters of our experiment, the R2 bidding strategy is given by

$$\beta_4^{R2}\left(s_i\right) = \begin{cases} 10 & \text{if} \quad s_i \leq 1 \\ 15 & \text{if} \quad s_i > 1 \end{cases}. \tag{A.20}$$

Thus, if $s_i \leq 1$, she will bid so low as to ensure she loses; if instead $s_i > 1$, she will bid

high enough to guarantee herself to win.

Next, consider a truthful level-0 (T0) player; this player bids the expected value of the prize conditional on just her signal:

$$\beta_4^{T0}(s_i) = \mathbb{E}[V|s_i] = 10 + s_i. \tag{A.21}$$

Notice that $\beta_4^{T0}(s_i) = \beta_4^{R1}(s_i)$.

A truthful level-1 (T1) player believes all of her opponents to be T0 players.In turn, her best response coincides with that of an R2 player; i.e., $\beta_4^{T1}(s_i) = \beta_4^{R2}(s_i)$.

A truthful level-2 (T2) player believes all of her opponents to be T1 players who bid according to β_4^{T1}. Given that T1 players bid either 10 or 15, a T2 player has three relevant options: i) bid strictly less than 10, guaranteeing herself a loss, ii) bid between 10 and 15, in which case she wins at a price of 10 but only if at least one of her opponents' signals is lower than 1, or iii) bid strictly more than 15, guaranteeing herself a win.Clearly, her expected payoff from bidding strictly less than 10 is equal to zero. Any bid between 10 and 15 yields the following expected payoff

$$\left[1 - \left(\frac{4}{5}\right)^4\right]\left(s_i + \frac{3178}{369} - 10\right) \tag{A.22}$$

where $s_i + \frac{3178}{369}$ denotes i's expectation of the value of the prize conditional on at least one of her opponents' signals being lower than 1.It is easy to see that expression (A.22) is strictly positive if and only if $s_i > \frac{512}{369}$.

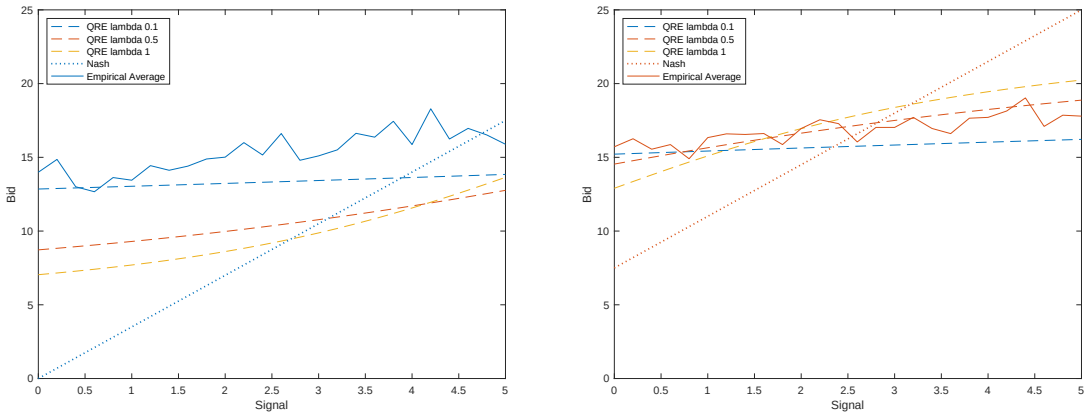
If she bids strictly more than 15, the bidder wins for sure and her expected payoff equals

$$\left[1 - \left(\frac{4}{5}\right)^4\right]\left(s_i + \frac{3178}{369} - 10\right) + \left(\frac{4}{5}\right)^4 [s_i + 4(3) - 15] \tag{A.23}$$

where $s_i + 4(3)$ denotes i's expectation of the value of the prize conditional on all of her opponents' signals being higher than 1. It is easy to verify that (A.23) exceeds (A.22) if and only if $s_i > 3$.

Hence, for the parameters of our experiment, the L2 bidding strategy is given by

$$\beta_4^{R2}(s_i) = \begin{cases} < 10 & \text{if} \quad s_i \leq \frac{512}{369} \\ \in [10, 15] & \text{if} \quad \frac{512}{369} < s_i \leq 3 \\ > 15 & \text{if} \quad 3 < s_i \end{cases}. \tag{A.24}$$



(a) One-Prize

(b) Four-Prize

Figure 15: Averages vs. Quantal Response Equilibrium

B.3 QRE

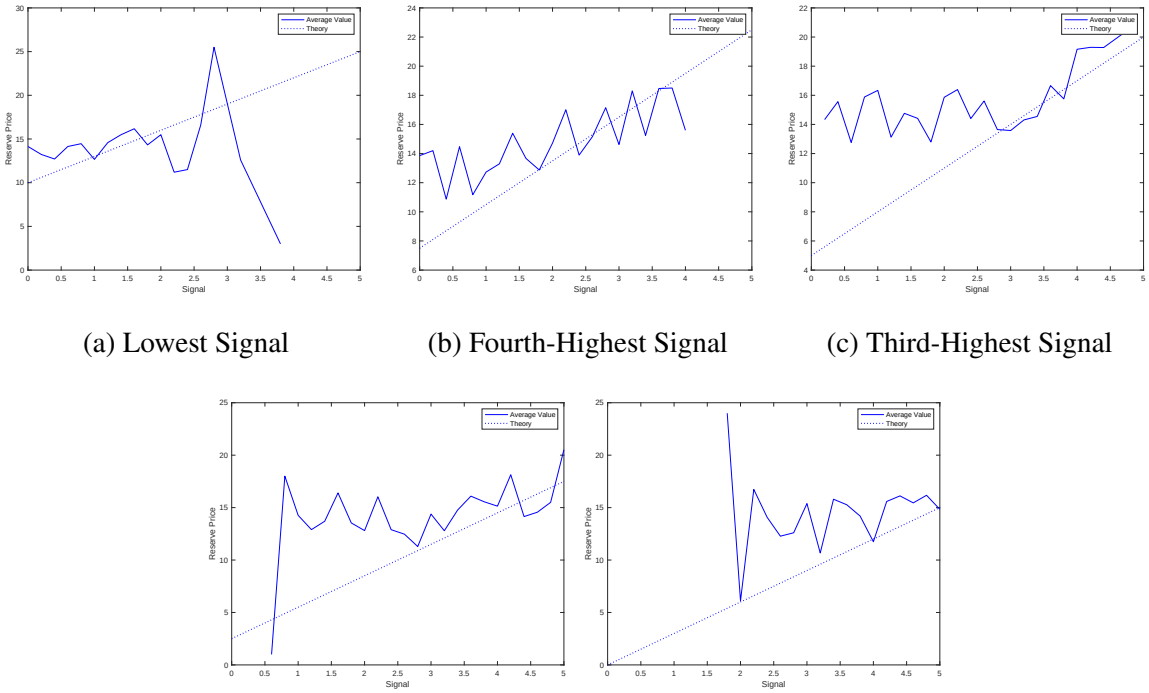
In this section, we present the results of calculating the distribution of bids under a Logistic Quantal Response Equilibrium (QRE) for various QRE parameters across different types of auctions. The computation of these probabilities is performed iteratively.

For each value of λ, we start by setting the bidding probabilities uniformly across all bids for each signal.[47] Assuming all players follow this initial distribution, we calculate the expected payoff for each possible bid. We then update the bidding probabilities by incorporating the expected payoffs and λ into the logit choice probabilities. This process continues until the bidding distribution stabilizes at a fixed point.

Figure 15 shows the average bidding functions generated by QRE for various values of λ. The empirical average of bidding in the one-prize auctions is higher than QRE's predicted average for almost all signals and values of λ, with the closest match being $\lambda = 0.1$. For the four-prize auctions, average bidding under $\lambda = 0.5$ matches many of the broad patterns of average empirical bids.

[47] Due to computational limitations, we restrict bids to dollar values instead of using increments of \$0.20 as in the actual experiment.

C Valuation Task Averages



(a) Lowest Signal (b) Fourth-Highest Signal (c) Third-Highest Signal

(d) Second-Highest Signal (e) Highest Signal

Figure 16: Average Valuation Choices vs. Risk-Neutral Benchmark, by Signal Rank

D AICs

Below, we report the AICs for each model both for estimating parameters separately across auctions (Sep) and restricting parameters to be the same across auctions (Same). We mark the model with the best AICs for the two estimation procedures with one or two asterisks, respectively.

ID	Crsd Sep	Lvl k Sep	QRE Sep	JoW Sep	Crsd Same	Lvl k Same	QRE Same	JoW Same
1	397.7	396.9	393.9	370.0*	400.1	398.4	398.9	382.7**
2	403.0	407.9	399.3*	401.9	399.0	402.2	397.3**	399.3
3	391.4	389.0*	389.4	391.2	395.7	394.9	394.7**	396.7
4	387.6	386.6	384.0	380.6*	388.9	388.6	387.9	385.4**
5	405.3	403.7	401.4	391.2*	401.6	402.7	399.7	388.4**
6	408.6	404.2	404.6	209.8*	405.4	407.4	403.4	260.8**
7	402.6	402.6	399.1	392.9*	401.6	400.2	400.1**	400.3
8	366.0	351.6*	362.6	366.5	362.3	348.7**	360.6	362.6
9	391.4	391.6	388.4*	388.9	392.9	397.0	390.9**	392.9
10	407.1	399.1	403.1	328.8*	405.4	406.9	403.4	365.8**
11	406.3	407.0	402.3	383.1*	405.4	407.3	403.4	380.1**
12	394.4	392.7	390.8	346.5*	397.7	395.2	396.7	382.4**
13	397.0	400.7	393.1*	396.7	394.7	397.5	392.7**	394.7
14	404.8	400.7	400.8	332.4*	405.4	407.2	403.4	335.3**
15	401.0	403.0	397.5*	401.4	397.1	397.5	395.6**	397.6
16	405.0	406.7	401.0*	404.8	401.3	403.5	399.3**	401.3
17	383.2	380.3	380.1	375.3*	385.0	383.5	384.6	381.3**
18	335.4	328.5*	340.2	335.2	341.0	336.6**	339.1	339.5
19	390.3	391.5	386.6*	387.2	386.4	385.8	384.7	383.4**
20	408.3	409.0	404.3	359.6*	405.4	407.4	403.4	360.0**
21	406.6	408.1	402.6	389.3*	405.4	407.4	403.4	386.8**
22	408.4	409.5	404.4	396.6*	405.4	407.4	403.4	394.2**
23	391.2	391.8	387.4*	391.0	387.3	393.0	385.5**	387.5
24	400.8	398.7	396.8	344.6*	404.4	402.4	402.8	363.6**
25	409.1	409.1	405.1	373.1*	405.4	407.4	403.4	369.9**
26	401.8	408.1	398.2	359.3*	405.4	407.4	403.4	384.0**
27	378.6	376.6	375.8	350.6*	375.4	370.6	375.4	362.4**
28	365.6	366.7	366.2	341.1*	405.4	407.4	403.4	393.9**
29	391.3	397.1	388.8*	389.2	388.9	392.7	388.3**	390.3
30	400.4	403.9	397.1	393.3*	402.9	403.9	401.3	398.5**
31	362.0	372.1	361.5	351.6*	363.2**	367.2	366.9	368.9
32	397.7	398.8	394.0*	397.1	398.8	401.8	396.9**	397.5
33	389.7	393.6	385.7	380.6*	405.4	406.4	403.4	394.4**
34	404.4	405.4	400.5	378.2*	402.4	404.2	400.6	376.9**
35	388.7	387.3	384.9	383.8*	387.5	386.7	386.3	383.6**
36	381.6	383.6	377.9	376.7*	388.8	389.5	388.0	386.1**
37	361.7	357.6*	360.8	361.0	362.9	356.4**	365.7	365.5
38	375.6	379.9	372.0*	372.2	372.7	381.9	370.7**	372.7
39	409.4	411.4	405.4*	409.3	405.4	407.4	403.4**	405.4
40	403.7	406.1	399.7	395.6*	400.0	402.1	398.1	393.3**

ID	Crsd Sep	Lvl k Sep	QRE Sep	JoW Sep	Crsd Same	Lvl k Same	QRE Same	JoW Same
41	408.7	407.7	404.7	377.6*	405.4	406.5	403.4	389.0**
42	397.8	394.8	393.8	378.0*	393.8	394.1	391.8	379.8**
43	408.5	400.7	404.5	304.7*	405.3	396.7	403.3	391.3**
44	396.7	392.1	393.3	381.8*	393.4	389.0	392.2	378.8**
45	383.4	389.0	382.5	377.6*	384.5**	385.7	385.3	387.3
46	405.0	405.9	401.0	396.5*	405.4	406.7	403.4	395.4**
47	397.2	401.6	393.2*	396.0	400.9	399.0	398.9**	400.9
48	403.1	408.6	399.1*	403.0	399.3	402.7	397.3**	399.3
49	408.1	411.6	404.2*	408.2	404.9	406.9	403.0**	405.0
50	380.3	377.2	377.1	374.2*	376.4	371.3**	375.6	377.6
51	345.5	359.5	342.3	342.1*	348.1	360.2	346.1**	348.1
52	360.0	360.8	356.8*	358.6	366.4	357.5**	364.4	364.8
53	395.9	392.0	392.6	371.2*	402.5	400.8	401.1	397.8**
54	403.6	401.6	399.6	365.4*	405.4	406.5	403.4	368.3**
55	406.7	406.3	402.7	342.4*	405.4	407.3	403.4	338.5**
56	398.1	392.0	394.7	341.5*	399.5	396.3	398.5	372.8**
57	404.8	402.5	400.9	396.8*	401.7	401.3	399.9**	401.7
58	406.1	398.2	402.1	381.0*	403.9	404.7	402.0	379.4**
59	303.2	299.0	373.5	237.3*	388.1	387.0	386.1	369.5**
60	396.2	393.8	392.4	377.4*	392.2	389.1	390.4	376.4**
61	404.6	387.3	400.6	365.7*	404.6	402.7	402.7	369.5**
62	310.4	331.0	306.4*	310.4	328.3	331.6	326.3**	328.3
63	391.2	388.7	387.5	382.6*	391.8	395.4	390.2	386.8**
64	406.9	403.5	402.9	376.4*	405.4	407.4	403.4	372.8**
65	394.4	389.6	390.4	364.7*	391.0	391.0	389.0	374.5**
66	409.4	412.4	405.4*	408.0	405.4	406.8	403.4**	404.1
67	403.9	405.1	399.9*	403.9	405.3	407.3	403.3**	405.3
68	408.7	406.2	404.7	324.2*	405.4	407.4	403.4	327.2**
69	380.9	393.3	376.9*	380.9	404.6	406.6	402.6**	404.6
70	363.5	359.7	360.6	334.3*	367.5	361.5	369.9	359.5**
71	375.0	361.1	373.4	324.0*	372.2	358.7	371.5	321.2**
72	396.8	401.5	392.8*	396.8	404.3	407.3	402.3**	404.3
73	399.6	400.2	395.8*	395.9	397.8	397.2	396.1	393.1**
74	394.1	391.2	390.1	377.8*	391.1	390.7	389.1	381.8**
75	394.4	404.3	390.9	374.4*	405.4	407.4	403.4	398.4**
76	405.0	403.2	401.0	351.0*	405.0	405.4	403.2	350.7**
77	307.3	289.4*	354.3	303.5	358.0	351.1	357.3	326.7**
78	363.3	363.9	362.4	360.4*	362.7	365.9	361.8**	363.8
79	396.9	396.5	393.6	381.2*	399.9	397.8	398.8	390.9**
80	407.8	410.7	403.9*	404.1	404.5	405.2	402.6**	403.3

ID	Crsd Sep	Lvl k Sep	QRE Sep	JoW Sep	Crsd Same	Lvl k Same	QRE Same	JoW Same
81	406.3	397.6	402.5	380.2*	403.6	403.3	401.8	385.9**
82	406.9	403.6	403.0	376.0*	403.2	403.2	401.4	375.4**
83	389.3	377.8	386.7	279.7*	390.5	389.7	388.5	366.6**
84	372.4	369.2	368.9*	372.9	374.0	369.6**	373.3	375.3
85	406.5	407.5	402.5	399.8*	405.3	406.1	403.3	397.5**
86	404.2	406.0	400.5*	404.0	400.5	401.4	398.7**	400.7
87	388.0	385.2	384.5	378.8*	393.3	393.6	392.1	386.1**
88	380.3	374.3	377.0	341.4*	378.1	371.5	377.7	356.1**
89	409.4	407.0	405.4*	406.3	405.4	402.5**	403.4	405.4
90	398.3	391.9	394.7	345.9*	401.8	398.7	400.5	366.6**
91	370.3	373.3	366.3	365.5*	372.8	371.5	370.8	369.2**
92	388.8	384.8	385.7	347.7*	386.7	383.1	386.2	363.2**
93	343.4	324.0*	342.4	332.4	339.4	320.8**	341.7	334.6
94	378.6	384.4	374.6*	378.6	398.9	402.2	396.9**	398.9
95	401.9	398.2	398.3	383.3*	399.1	398.9	397.6**	399.6
96	298.6*	299.4	306.4	305.2	331.2	311.3**	337.1	339.1
97	407.4	398.9	403.4	354.1*	405.1	406.4	403.1	384.5**
98	407.5	407.2	403.5	365.5*	405.4	407.2	403.4	363.8**
99	404.6	398.7	400.7	382.4*	401.7	402.9	399.8	382.8**
100	402.8	403.5	398.8	369.5*	405.4	406.0	403.4	384.9**
101	383.1	393.0	379.1*	383.1	398.5	400.7	396.5**	398.5
102	377.2	371.1	375.0	366.5*	384.2	381.6**	384.5	382.1
103	399.6	401.0	395.7	374.0*	402.7	404.8	401.1	383.7**
104	401.5	394.1	398.0	348.6*	397.6	395.5	396.0	360.7**
105	409.4	410.8	405.4	215.6*	405.4	407.4	403.4	229.2**
106	406.6	401.1	402.6	379.3*	405.4	403.8	403.4	375.3**
107	399.3	399.0	395.4	383.3*	402.0	403.3	400.4	385.4**
108	405.7	407.6	402.0	373.1*	402.9	402.0	401.3**	403.3
109	302.6	284.2*	331.3	326.7	321.7	296.9**	330.2	332.2
110	393.2	387.9	389.3	367.0*	399.6	400.5	398.2	379.6**
111	404.8	403.3	400.8	368.9*	405.4	407.4	403.4	369.3**
112	404.8	402.4	400.9	378.7*	402.1	402.9	400.4	375.2**
113	390.3	387.8	387.2*	388.3	405.4	407.4	403.4**	405.4
114	363.0	368.3	359.4*	360.0	367.4	366.3**	367.9	369.9
115	369.9	368.6	366.8*	370.8	399.8	400.0	398.1**	399.9
116	390.7	394.2	386.9*	389.8	391.9	393.1	390.5**	392.3
117	368.3	382.3	364.9*	368.0	368.8	382.2	366.8**	368.8
118	390.5	390.5	387.1	384.4*	395.6	398.6	394.4	389.9**
119	382.0	370.1	379.1	365.3*	379.5	375.5	377.5	370.6**
120	408.0	404.2	404.0	397.4*	404.7	399.7	402.7	399.4**

E Experimental Screenshots

Below, we include screenshots from the experiment.

Introduction

PLEASE READ CAREFULLY AND DO NOT PRESS NEXT UNTIL INSTRUCTED TO DO SO.

Thank you for participating in this study. This study is about decision-making. It should take about 90-120 minutes, and you will be paid based on your earnings from the experiment. The money you earn will be paid either in cash at the end of the study or electronically within a few days of the end of the study.

Please do not use any electronic devices or talk with other participants during this study.

There will be no deception in this study. Every decision you make will be carried out exactly as it is described in the instructions. Anything else would violate the human ethics protocol under which we run the study (UQ Human Research Ethics Approval 2023/HE001894).

In the study you will make decisions that will affect the amount of money you earn. The study will consist of games that you will play with other randomly selected players. The players that you are paired with in a match are selected independently of who you play with in any other match.

Please pay close attention to the instructions on the next page. After you read these instructions, there will be a short quiz on the instructions. You will receive $1 for each question you answer correctly.

If you have questions at any point, please raise your hand and we will answer your questions privately.

Next

Figure 17: Introduction

Instructions

PLEASE READ CAREFULLY AND DO NOT PRESS NEXT UNTIL INSTRUCTED TO DO SO.

AFTER THESE INSTRUCTIONS THERE WILL BE A SHORT QUIZ ABOUT THESE INSTRUCTIONS. YOU WILL RECEIVE $1 FOR EACH QUESTION YOU ANSWER CORRECTLY.

All participants will receive a show-up-fee of $15 regardless of what happens during the study.

In this study, you will participate in 20 rounds of decision-making. Each round begins with the computer randomly pairing you with four other participants. The players that you are paired with in a round are selected independently of whom you play with in any other round. You will not know the players that you are paired with.

In each round, you will have the chance to win a prize. The value of the prize is random and will change from round to round. While no player will know the value of the prize, each player will receive a signal of the value. Each signal is drawn independently and has an equal chance of taking each value between $0 and $5 (in $0.2 increments). The value of the prize is the **SUM** of all 5 players' signals.

Each round is divided into two stages: the bidding stage and the purchase stage.

Next

Figure 18: Basic Instructions

Instructions: Bidding Stage

PLEASE READ CAREFULLY AND DO NOT PRESS NEXT UNTIL INSTRUCTED TO DO SO.

AFTER THESE INSTRUCTIONS THERE WILL BE A SHORT QUIZ ABOUT THESE INSTRUCTIONS. YOU WILL RECEIVE $1 FOR EACH QUESTION YOU ANSWER CORRECTLY.

In the bidding stage of each round, you will participate in two different types of auctions. We refer to them as the "one-prize auction" and the "four-prize auction". Before making bids in these auctions, all players will be told the value of their own signal (but will not be told anything about the other players' signals).

In the **one-prize auction**, all five players receive a bidding budget of $30 from which they make bids. The winner of the auction will be the player with the highest bid, with any ties broken randomly amongst the players with equally high bids. The winner receives the prize and pays a price equal to the second-highest bid. So their total payoff is the budget, plus the value of the prize, minus the price. All of the other players keep their full budget.

In the **four-prize auction**, all five players receive a bidding budget of $30 from which they make bids. The four highest bidders will win the auction, with any ties broken randomly amongst the players tied for the fourth highest bid. The winners receive the prize and pay a price equal to the fifth-highest bid. So their total payoff is the budget, plus the value of the prize, minus the price. The player that does not win keeps their full budget.

Next

Figure 19: Instructions: Bidding Stage

Instructions: Purchase Stage

PLEASE READ CAREFULLY AND DO NOT PRESS NEXT UNTIL INSTRUCTED TO DO SO.

AFTER THESE INSTRUCTIONS THERE WILL BE A SHORT QUIZ ABOUT THESE INSTRUCTIONS. YOU WILL RECEIVE $1 FOR EACH QUESTION YOU ANSWER CORRECTLY.

In the purchase stage of each round, you will be asked whether you would be willing to purchase one of the prizes from the bidding stage at various prices. No player's signal will change, and the value of the prize will still be equal to the sum of those signals. However, before making your purchase decisions you will be told how your signal ranks against other players' signals. For instance, if your signal is $3.2 and you are told it is the fourth highest, that means that one player has a signal that is no higher than $3.2 and three players have signals that are no lower than $3.2.

You will be asked for the maximum price at which you would be willing to purchase the prize. After you choose your maximum price, a random price will be drawn and compared to your maximum price. If the randomly drawn price is (weakly) lower than your maximum price, then you will purchase a prize at the randomly drawn price out of your budget of $30. Otherwise you will keep your full budget of $30, but will not receive a prize.

The price that you end up facing will be chosen randomly between $0.2 and $30, with each multiple of $0.2 being equally likely to be drawn. Your choices cannot affect the price that is selected. Therefore, it is in your best interest to answer each question as if you were facing that price with certainty.

Important: Your outcome from the purchase stage will not depend on what other players choose. The price that is implemented will be chosen randomly from all prices, and will not depend on what other players choose. Furthermore, there are no restrictions on the number of players that can receive prizes: it is possible that everyone receives a prize, but it is also possible that no one receives a prize.

Next

Figure 20: Instructions: Purchase Stage

Instructions: Payoffs

PLEASE READ CAREFULLY AND DO NOT PRESS NEXT UNTIL INSTRUCTED TO DO SO.

AFTER THESE INSTRUCTIONS THERE WILL BE A SHORT QUIZ ABOUT THESE INSTRUCTIONS. YOU WILL RECEIVE $1 FOR EACH QUESTION YOU ANSWER CORRECTLY.

At the end of the study, one round will be chosen at random to be the one that counts. Within the round that counts, either the **one-prize auction**, the **four-prize auction**, or the **purchase stage** will be chosen to determine your payoffs, each with a one-in-three chance.

On the next page, you will find an example round to help you better understand the rules. Please go through it at your own pace. After you have reviewed the example, click next again to begin the quiz.

Next

Figure 21: Instructions: Payoffs

Example

PLEASE READ THE EXAMPLE CAREFULLY AND CLICK NEXT WHEN YOU ARE READY TO BEGIN THE QUIZ.

The table below shows an example of the players' (A, B, C, D, and E) signals and decisions in each task.

Player	Signal	One-Prize Auction Bid	Four-Prize Auction Bid	Purchase Stage Choice
A	$1.40	$4.80	$9.00	$3.40
B	$0.80	$3.60	$3.80	$2.80
C	$4.00	$23.00	$12.40	$20.00
D	$1.40	$9.20	$9.60	$6.40
E	$3.40	$9.00	$8.20	$9.80

In this example, the value of the prize (for all players and every decision) is:
$1.40+$0.80+$4.00+$1.40+$3.40=$11.

In the **one-prize auction**, Player C is the winner as she bid the highest; hence, she gets the prize and pays $9.20 for it, as this was the value of the second-highest bid. Player C's payoff from this auction is $30+$11-$9.20=$31.80, while all other players' payoffs are $30.

In the **four-prize auction**, Players A, C, D and E are the four highest bidders; hence, they each get the prize and pay $3.80 for it, as this was the value of the lowest bid. The four winners' payoffs are $30+$11-$3.80=$37.20, while Player B's payoff is $30.

For the **purchase stage**, suppose that the randomly drawn price is $4. Consider first Player B who was told that she had the lowest signal and indicated that she was willing to purchase the prize at any price lower than or equal to $2.80. In this case, given that the randomly drawn price is higher than her maximum purchase price, she would not purchase the prize and she would receive a payoff of $30. Next, consider Player E who was told that he had the second-highest signal and indicated that he was willing to purchase the prize at any price lower than or equal to $9.80. In this case, given that the randomly drawn price is lower than his maximum purchase price, he would purchase the prize at a price of $4 and he would receive a payoff of $30+$11-$4=$37.

Next

Figure 22: Example

Quiz

You will now be given a series of questions to check your understanding of the instructions and examples. You will be paid $1 for each answer you get correct.

Suppose that your signal is $2.2, and the other four signals are $1.6, $3.4, $4.8, and $5. What is the value of the prize?

- $11
- $13
- $15
- $17

How many players are in each group?

- Two.
- Three.
- Four.
- Five.

Suppose that in the one-prize auction, you bid $8 and the other bids are $4, $5, $11, and $12. What is true about the outcome?

- You win a prize. The price in the auction is $11.
- You win a prize. The prize in the auction is $4.
- You do not win a prize. The price in the auction is $11.
- You do not win a prize. The price in the auction is $4.

Suppose that in the four-prize auction, you bid $8 and the other bids are $4, $5, $11, and $12. What is true about the outcome?

- You win a prize. The price in the auction is $11.
- You win a prize. The prize in the auction is $4.
- You do not win a prize. The price in the auction is $11.
- You do not win a prize. The price in the auction is $4.

Suppose that in the one-prize auction, you bid $12 and the other bids are $5, $6, $10, and $16. What is true about the outcome?

- You win a prize. The price in the auction is $12.
- You win a prize. The price in the auction is $5.
- You do not win a prize. The price in the auction is $12.
- You do not win a prize. The price in the auction is $5.

What is true about how the price is determined in the purchase stage?

- The price that is selected will be higher if the value of the prize is higher.
- Each possible value of the price is equally likely to be chosen.
- Another player will choose your price.
- The price is determined by the outcome of the four-prize auction.

What is true about the how prizes are distributed in the purchase stage?

- Either everyone wins the prize, or no one does.
- One player in each group will win a prize.
- Four players in each group will win a prize.
- Other players' choices do not affect your chances to receive a prize.

When you believe you have answered all questions correctly, press next to check your answers.

Next

Figure 23: Quiz

Quiz Answers

The answers for the quiz are given below. Please review the answers and note any mistakes you have made.

Question 1: Suppose that your signal is $2.2, and the other four signals are $1.6, $3.4, $4.8, and $5. What is the value of the prize?
Correct Answer: $17
Your Answer: $15

Question 2: How many players are in each group?
Correct Answer: Five.
Your Answer: Four.

Question 3: Suppose that in the one-prize auction, you bid $8 and the other bids are $4, $5, $11, and $12. What is true about the outcome?
Correct Answer: You do not win a prize. The price in the auction is $11.
Your Answer: You do not win a prize. The price in the auction is $11.

Question 4: Suppose that in the four-prize auction, you bid $8 and the other bids are $4, $5, $11, and $12. What is true about the outcome?
Correct Answer: You win a prize. The prize in the auction is $4.
Your Answer: You do not win a prize. The price in the auction is $11.

Question 5: Suppose that in the one-prize auction, you bid $12 and the other bids are $5, $6, $10, and $16. What is true about the outcome?
Correct Answer: You do not win a prize. The price in the auction is $12.
Your Answer: You do not win a prize. The price in the auction is $12.

Question 6: What is true about how the price is determined in the purchase stage?
Correct Answer: Each possible value of the price is equally likely to be chosen.
Your Answer: Another player will choose your price.

Question 7: What is true about the how prizes are distributed in the purchase stage?
Correct Answer: Other players' choices do not affect your chances to receive a prize.
Your Answer: Four players in each group will win a prize.

You earned $2.0 from your correct answers. Please review any questions you answered incorrectly. When you are ready to begin the first round, click the next button.

Next

Figure 24: Quiz Answers

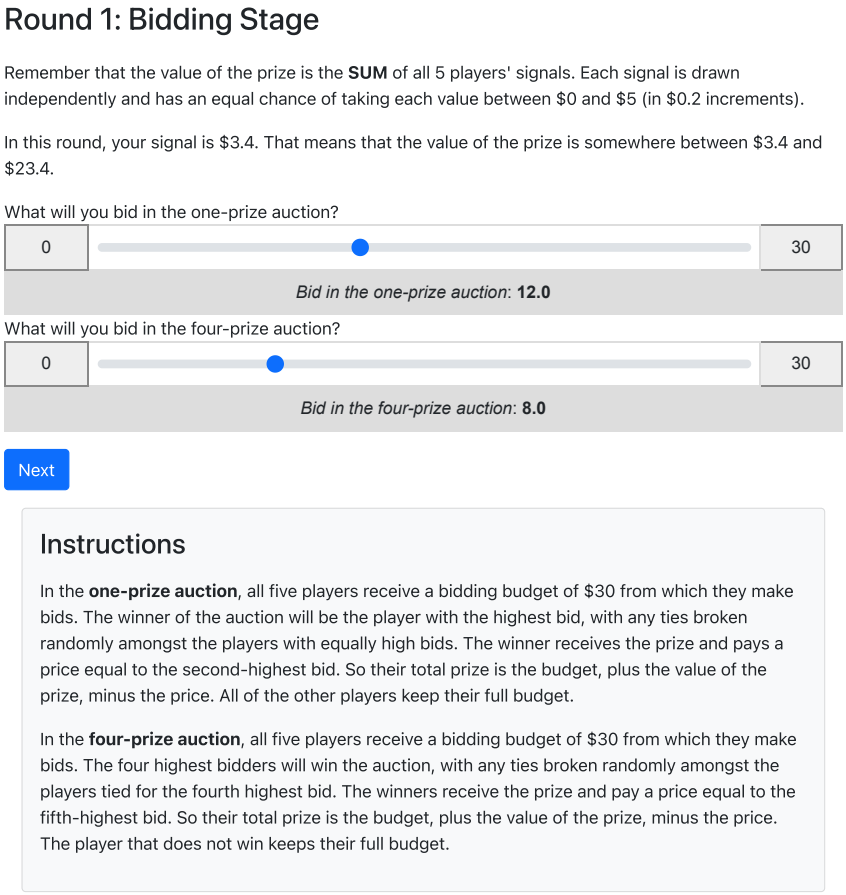


Figure 25: Bidding Stage

Round 1: Purchase Stage

Remember that the value of the prize is the **SUM** of all 5 players' signals. Each signal is drawn independently and has an equal chance of taking each value between $0 and $5 (in $0.2 increments).

In this round, your signal is $3.4. That means that the value of the prize is somewhere between $3.4 and $23.4.

We can also inform you that **your signal is the second highest of the five signals**. That means that one player in your group has a signal that is greater than or equal to yours and three players in your group have signals that are less than or equal to yours.

What is the maximum price at which you would purchase the prize?

0 — 30

Maximum price at which you will buy: **17.2**

Next

Instructions

You will be asked for the maximum price at which you would be willing to purchase the prize. After you choose your maximum price, a random price will be drawn and compared to your maximum price. If the randomly drawn price is (weakly) lower than your maximum price, then you will purchase a prize at the randomly drawn price out of your budget of $30. Otherwise you will keep your full budget of $30, but will not receive a prize.

The price that you end up facing will be chosen randomly between $0.20 and $30, with each multiple of $0.20 being equally likely to be drawn. Your choices cannot affect the price. Therefore, it is in your best interest to answer each question as if you were facing that price with certainty.

Price	Purchase?	Price	Purchase?	Price	Purchase?	Price	Purchase?	Price	Purchase?	Price	Purchase?
$0.20	✔	$5.20	✔	$10.20	✔	$15.20	✔	$20.20	✗	$25.20	✗
$0.40	✔	$5.40	✔	$10.40	✔	$15.40	✔	$20.40	✗	$25.40	✗
$0.60	✔	$5.60	✔	$10.60	✔	$15.60	✔	$20.60	✗	$25.60	✗
$0.80	✔	$5.80	✔	$10.80	✔	$15.80	✔	$20.80	✗	$25.80	✗
$1.00	✔	$6.00	✔	$11.00	✔	$16.00	✔	$21.00	✗	$26.00	✗
$1.20	✔	$6.20	✔	$11.20	✔	$16.20	✔	$21.20	✗	$26.20	✗
$1.40	✔	$6.40	✔	$11.40	✔	$16.40	✔	$21.40	✗	$26.40	✗
$1.60	✔	$6.60	✔	$11.60	✔	$16.60	✔	$21.60	✗	$26.60	✗
$1.80	✔	$6.80	✔	$11.80	✔	$16.80	✔	$21.80	✗	$26.80	✗
$2.00	✔	$7.00	✔	$12.00	✔	$17.00	✔	$22.00	✗	$27.00	✗
$2.20	✔	$7.20	✔	$12.20	✔	$17.20	✔	$22.20	✗	$27.20	✗
$2.40	✔	$7.40	✔	$12.40	✔	$17.40	✗	$22.40	✗	$27.40	✗
$2.60	✔	$7.60	✔	$12.60	✔	$17.60	✗	$22.60	✗	$27.60	✗
$2.80	✔	$7.80	✔	$12.80	✔	$17.80	✗	$22.80	✗	$27.80	✗
$3.00	✔	$8.00	✔	$13.00	✔	$18.00	✗	$23.00	✗	$28.00	✗
$3.20	✔	$8.20	✔	$13.20	✔	$18.20	✗	$23.20	✗	$28.20	✗
$3.40	✔	$8.40	✔	$13.40	✔	$18.40	✗	$23.40	✗	$28.40	✗
$3.60	✔	$8.60	✔	$13.60	✔	$18.60	✗	$23.60	✗	$28.60	✗
$3.80	✔	$8.80	✔	$13.80	✔	$18.80	✗	$23.80	✗	$28.80	✗
$4.00	✔	$9.00	✔	$14.00	✔	$19.00	✗	$24.00	✗	$29.00	✗
$4.20	✔	$9.20	✔	$14.20	✔	$19.20	✗	$24.20	✗	$29.20	✗
$4.40	✔	$9.40	✔	$14.40	✔	$19.40	✗	$24.40	✗	$29.40	✗
$4.60	✔	$9.60	✔	$14.60	✔	$19.60	✗	$24.60	✗	$29.60	✗
$4.80	✔	$9.80	✔	$14.80	✔	$19.80	✗	$24.80	✗	$29.80	✗
$5.00	✔	$10.00	✔	$15.00	✔	$20.00	✗	$25.00	✗	$30.00	✗

Figure 26: Purchase Stage

Round 1: Results

We can now inform you that the value of the prize was $13.8.

One-prize auction: You were not the winner. Your bid was $12 and the price (the second highest bid) was $12. So your payoff from the one-prize auction is $30.

Four-prize auction: You were one of the winners. Your bid was $8 and the price (the lowest bid) was $1.2. So your payoff from the four-prize auction is $30 + $13.8 - $1.2 = $42.6.

Purchase stage: Your randomly selected price was $0.4. The maximum price you indicated that you were willing to pay was $17.2, which is at least as high as the price. Thus, you purchased the prize and your payoff from the purchase stage is $30 + $13.8 - $0.4 = $43.4.

Overall, if this round is chosen to be the one that counts, you are equally likely to receive your payoff from the one-prize auction ($30), the four-prize auction ($42.6), and the purchase stage ($43.4).

Next

Figure 27: Results